

Next Step/Workup Questions

Category: Next Step/Workup **Total Questions:** 206 **Purpose:** Medical professional review of categorization accuracy

Review Instructions

For each question below, please mark:

- ✓ **Correctly categorized** - Question clearly belongs in this category
- ✗ **Miscategorized** - Question belongs in a different category (note which one)
- ? **Ambiguous** - Question could reasonably fit multiple categories

You don't need to review all questions! Even 20-30 per category is very helpful.

What 'Next Step/Workup' Means

Questions asking what test to order, what action to take next, or how to proceed with patient evaluation.

Questions for Review

Question 1

Clinical Scenario:

An otherwise healthy, exclusively breastfed 4-day-old neonate is brought to the physician because of yellowing of his skin and eyes. His urine has been clear and stools have been normal. He was born at term by vacuum-assisted delivery and weighed 4000 g (8 lb 8 oz). Pregnancy was complicated by gestational diabetes mellitus. His older sibling had jaundice in the neonatal period. Vital signs are within normal limits. He appears alert and comfortable. Physical examination shows jaundice of the skin and sclerae. The liver is palpated 1 cm below the right costal margin. Laboratory studies show: Hemoglobin 17 g/dl Reticulocyte count 0.5 % Total bilirubin 21.2 mg/dl Direct bilirubin 2 mg/dl Indirect bilirubin 19.1 mg/dl Coombs test Negative Which of the following is the most appropriate next step in management?"

Answer Choices:

A. Intravenous immunoglobulin B. Increase frequency of breast feeds C. MRI of the brain D. Phototherapy

Correct Answer: D. Phototherapy

Question 2

Clinical Scenario:

A 30-year-old woman, gravida 2, para 1, at 40 weeks' gestation is admitted to the hospital in active labor. Pregnancy has been complicated by iron deficiency anemia, which was treated with iron supplements. Her first pregnancy and vaginal delivery were uncomplicated. There is no personal or family history of serious illness. Her pulse is 90/min, respirations are 15/min, and blood pressure is 130/80 mm Hg. The abdomen is nontender and contractions are felt. Ultrasonography shows that the fetal long axis is at a right angle compared to the long axis of the maternal uterus. The fetal heart rate is 140/min and is reactive with no decelerations. Which of the following is the most appropriate next step in the management of this patient?

Answer Choices:

A. Administration of oxytocin and normal vaginal birth B. Lateral positioning of the mother C. Cesarean section D. External cephalic version

Correct Answer: C. Cesarean section

Question 3

Clinical Scenario:

A 53-year-old homeless woman is brought to the emergency department by the police after she was found in the park lying unconscious on the ground. Both of her pupils are normal in size and reactive to light. There are no signs of head trauma. Finger prick test shows a blood glucose level of 20 mg/dL. She has been brought to the emergency department for acute alcohol intoxication several times before. Her vitals signs include: blood pressure 100/70 mm Hg, heart rate 90/min, respiratory rate 22/min, and temperature 35.0°C (95.0°F). On general examination, she looks pale, but there is no sign of icterus noted. On physical examination, the abdomen is soft and non-tender and no hepatosplenomegaly noted. She spontaneously opens her eyes after the administration of a bolus of intravenous dextrose, thiamine, and naloxone. Blood and urine samples are drawn for toxicology screening. Finally, the blood alcohol level turns out to be 300 mg/dL. What will be the most likely laboratory findings in this patient?

Answer Choices:

A. AST > ALT, increased gamma-glutamyl transferase B. Decreased ALP C. AST > ALT, normal gamma glutamyl transferase D. Decreased MCV

Correct Answer: A. AST > ALT, increased gamma-glutamyl transferase

Question 4

Clinical Scenario:

A 32-year-old man comes to the physician because of a 2-week history of a cough and shortness of breath. He also noted several episodes of blood-tinged sputum over the last 4 days. He has a 3-month history of progressive fatigue. His temperature is 37.5°C (98.6°F), pulse is 86/min, respirations are 17/min, and blood pressure is 150/93 mm Hg. Examination shows pale conjunctivae. Crackles are heard on auscultation of the chest. Laboratory studies show: Hemoglobin 10.2 g/dL Leukocyte count 9200/mm³ Platelet count 305,000/mm³ Serum Na⁺ 136 mEq/L Cl⁻ 101 mEq/L K⁺ 4.5 mEq/L HCO₃⁻ 25 mEq/L Urea nitrogen 28 mg/dL Creatinine 2.3 mg/dL Anti-GBM antibodies positive Antinuclear antibodies negative Urine Blood 2+ Protein 2+ RBC 11–13/hbf RBC casts rare He is started on prednisone and cyclophosphamide. Which of the following is the most appropriate next step in management?"

Answer Choices:

A. Administer inhalative fluticasone B. Perform hemodialysis C. Perform plasmapheresis D. Administer enalapril

Correct Answer: C. Perform plasmapheresis

Question 5

Clinical Scenario:

A 57-year-old man comes to the physician with a 3-month history of right flank pain. Urinalysis shows 60 RBC/hpf. Renal ultrasound shows a 3 cm, well-defined mass in the upper pole of the right kidney. A photomicrograph of a section of the resected mass is shown. Which of the following is the most likely diagnosis?

Answer Choices:

A. Clear cell renal carcinoma B. Oncocytoma C. Nephroblastoma D. Angiomyolipoma

Correct Answer: B. Oncocytoma

Question 6

Clinical Scenario:

A 67-year-old Caucasian female presents to her primary care physician after a screening DEXA scan reveals a T-score of -3.0. Laboratory work-up reveals normal serum calcium, phosphate, vitamin D, and PTH levels. She smokes 1-2 cigarettes per day. Which of the following measures would have reduced this patient's risk of developing osteoporosis?

Answer Choices:

A. Reduced physical activity to decrease the chance of a fall B. Initiating a swimming exercise program three days per week C. Calcium and vitamin D supplementation D. Weight loss

Correct Answer: C. Calcium and vitamin D supplementation

Question 7

Clinical Scenario:

A 1-month-old infant is brought to the physician for a well-child examination. His mother reports that she had previously breastfed her son every 2 hours for 15 minutes but is now feeding him every 4 hours for 40 minutes. She says that the infant sweats a lot and is uncomfortable during feeds. He has 6 wet diapers and 2 stools daily. He was born at 36 weeks' gestation. He currently weighs 3500 g (7.7 lb) and is 52 cm (20.4 in) in length. He is awake and alert. His temperature is 37.1°C (98.8°F), pulse is 170/min, respirations are 55/min, and blood pressure is 80/60 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 99%. Cardiopulmonary examination shows a 4/6 continuous murmur along the upper left sternal border. After confirming the diagnosis via echocardiography, which of the following is the most appropriate next step in the management of this patient?

Answer Choices:

A. Prostaglandin E1 infusion B. Percutaneous surgery C. Digoxin and furosemide D. Indomethacin infusion

Correct Answer: D. Indomethacin infusion

Question 8

Clinical Scenario:

A 64-year-old nulliparous woman comes to the physician because of fatigue and an increase in abdominal girth despite a 5-kg (11.0-lb) weight loss over the past 6 months. Her last Pap smear 2 years ago showed atypical squamous cells of undetermined significance; subsequent HPV testing was negative at that time. Menarche was at the age of 10 years and her last menstrual period was 6 years ago. Abdominal examination shows shifting dullness. There is tenderness to palpation of the left lower quadrant but no guarding or rebound. Bimanual palpation shows a small uterus and a left adnexal mass. Further evaluation of this patient is most likely to show which of the following findings?

Answer Choices:

A. Proliferation of endometrial glands B. Chocolate cyst of the left ovary C. Elevated serum CA-125 level D. Cervical dysplasia on cervical smear

Correct Answer: C. Elevated serum CA-125 level

Question 9

Clinical Scenario:

A 46-year-old man accountant is admitted to the emergency department with complaints of retrosternal crushing pain that radiates to his left arm and jaw. The medical history is significant for hyperlipidemia and arterial hypertension, for which he is prescribed a statin and ACE inhibitor, respectively. An ECG is obtained and shows an ST-segment elevation in leads avF and V2-V4. The blood pressure is 100/50 mm Hg, the pulse is

120/min, and the respiratory rate is 20/min. His BMI is 33 kg/m² and he has a 20-year history of smoking cigarettes. Troponin I is elevated. The patient undergoes percutaneous coronary intervention immediately after admission. Angioplasty and stenting were successfully performed. On follow-up the next day, the ECG shows decreased left ventricular function and local hypokinesia. The patient is re-evaluated 14 days later. The echocardiography reveals a normal ejection fraction and no hypokinesis. Which of the phenomena below explains the patient's clinical course?

Answer Choices:

A. Coronary steal syndrome B. Reperfusion injury C. Myocardial stunning D. Coronary collateral circulation

Correct Answer: C. Myocardial stunning

Question 10

Clinical Scenario:

A 13-year-old boy is brought to the emergency department because of a 2-day history of fever, headache, and irritability. He shares a room with his 7-year-old brother, who does not have any symptoms. The patient appears weak and lethargic. His temperature is 39.1°C (102.4°F) and blood pressure is 99/60 mm Hg. Physical examination shows several purple spots over the trunk and extremities. A lumbar puncture is performed. Gram stain of the cerebrospinal fluid shows numerous gram-negative diplococci. Administration of which of the following is most likely to prevent infection of the patient's brother at this time?

Answer Choices:

A. Penicillin G B. Cephalexin C. Rifampin D. Doxycycline

Correct Answer: C. Rifampin

Question 11

Clinical Scenario:

A 5-month-old boy is brought to the physician by his mother because of poor weight gain and chronic diarrhea. He has had 3 episodes of otitis media since birth. Pregnancy and delivery were uncomplicated but his mother received no prenatal care. His immunizations are up-to-date. He is at the 10th percentile for height and 5th percentile for weight. Physical examination shows thick white plaques on the surface of his tongue that can be easily scraped off with a tongue blade. Administration of which of the following is most likely to have prevented this patient's condition?

Answer Choices:

A. Fluconazole B. Penicillin G C. Zidovudine D. Rifampin

Correct Answer: C. Zidovudine

Question 12

Clinical Scenario:

A 57-year-old male with diabetes mellitus type II presents for a routine check-up. His blood glucose levels have been inconsistently controlled with medications and diet since his diagnosis 3 years ago. At this current visit, urinalysis demonstrates albumin levels of 250 mg/day. All prior urinalyses have shown albumin levels below 20 mg/day. At this point in the progression of the patient's disease, which of the following is the most likely finding seen on kidney biopsy?

Answer Choices:

A. Normal kidney biopsy, no pathological finding is evident at this time
B. Glomerular basement membrane thickening and mesangial expansion
C. Kimmelstiel-Wilson nodules and tubulointerstitial fibrosis
D. Significant global glomerulosclerosis

Correct Answer: B. Glomerular basement membrane thickening and mesangial expansion

Question 13

Clinical Scenario:

A 29-year-old Mediterranean man presents to the clinic for fatigue and lightheadedness for the past week. He reports an inability to exercise as his heart would beat extremely fast. He was recently diagnosed with active tuberculosis and started on treatment 2 weeks ago. He denies fever, weight loss, vision changes, chest pain, dyspnea, or bloody/dark stools. A physical examination is unremarkable. A peripheral blood smear is shown in figure A. What is the most likely explanation for this patient's symptoms?

Answer Choices:

A. Abnormally low level of glutathione activity
B. Drug-induced deficiency in vitamin B6
C. Inhibition of ferrochelatase and ALA dehydratase
D. Iron deficiency

Correct Answer: B. Drug-induced deficiency in vitamin B6

Question 14

Clinical Scenario:

A 47-year-old woman is upset with her neighbor for playing music too loudly late at night. Rather than confront her neighbor directly, the woman makes a habit of parking her car in a manner that makes it difficult for her neighbor to park in his spot. Which of the following defense mechanisms is this woman demonstrating?

Answer Choices:

A. Sublimation
B. Regression
C. Acting out
D. Passive aggression

Correct Answer: D. Passive aggression

Question 15

Clinical Scenario:

A 63-year-old man presents to the emergency room because of worsening breathlessness that began overnight. He was diagnosed with asthma 3 years ago and has been using albuterol and steroid inhalers. He does not have a prior history of cardiac disease or other respiratory diseases. The man is a retired insurance agent and has lived his entire life in the United States. His vital signs include: respiratory rate 40/min, blood pressure 130/90 mm Hg, pulse rate 110/min, and temperature 37.0°C (98.6°F). Physical examination shows severe respiratory distress, with the patient unable to lie down on the examination table. Auscultation of the chest reveals widespread wheezes in the lungs and the presence of S3 gallop rhythm. The man is admitted to hospital and laboratory investigations and imaging studies are ordered. Test results include the following: WBC count $18.6 \times 10^9/L$ Eosinophil cell count $7.6 \times 10^9/L$ (40% eosinophils) Troponin T 0.5 ng/mL Anti-MPO (P-ANCA) antibodies positive Anti-PR3-C-ANCA negative Immunoglobulin E 1,000 IU/mL Serological tests for HIV, echovirus, adenovirus, Epstein-Barr virus, and parvovirus B19 are negative. ECG shows regular sinus tachycardia with an absence of strain pattern or any evidence of ischemia. Transthoracic echocardiography reveals a dilated left ventricle with an ejection fraction of 30% (normal is 55% or greater). Which of the following diagnoses best explains the clinical presentation and laboratory findings in this patient?

Answer Choices:

A. Eosinophilic granulomatosis with polyangiitis (EGPA) B. Chagas disease C. Primary dilated cardiomyopathy D. Extrinsic asthma

Correct Answer: A. Eosinophilic granulomatosis with polyangiitis (EGPA)

Question 16

Clinical Scenario:

A 57-year-old construction worker presents with gradually worsening shortness of breath for the past several months and left pleuritic chest pain for 2 weeks. He denies fever, cough, night sweats, wheezing, or smoking. He is recently diagnosed with hypertension and started amlodipine 10 days ago. He has been working in construction for the last 25 years and before that, he worked at a ship dry-dock for 15 years. Physical exam reveals bilateral crackles at the lung bases. Chest X-ray reveals bilateral infiltrates at the lung bases. Pulmonary function tests show a slightly increased FEV1/FVC ratio, but total lung volume is decreased. CT scan shows pleural scarring. What of the following conditions is the most likely explanation in this case?

Answer Choices:

A. Asbestosis B. Drug-induced interstitial lung disease C. Sarcoidosis D. Allergic bronchopulmonary aspergillosis

Correct Answer: A. Asbestosis

Question 17

Clinical Scenario:

A 34-year-old G3P2103 with a past medical history of preeclampsia in her last pregnancy, HIV (CD4: 441/mm³), and diabetes mellitus presents to her obstetrician for her first postpartum visit. She delivered her third child via C-section one week ago and reports that she is healing well from the surgery. She says that breastfeeding has been going well and that her baby has nearly regained his birth weight. The patient complains that she has been more tired than expected despite her efforts to sleep whenever her baby is napping. She relies on multiple iced coffees per day and likes to eat the ice after she finishes the drink. Her diet is otherwise unchanged, and she admits that she has not been getting outside to exercise as much as usual. Her home medications include metformin and her HAART regimen of dolutegravir, abacavir, and lamivudine. Her temperature is 98.9°F (37.2°C), blood pressure is 128/83 mmHg, pulse is 85/min, and respirations are 14/min. On physical exam, she is tired-appearing with conjunctival pallor.

This patient is at risk of developing which of the following conditions?

Answer Choices:

A. Hemolytic anemia B. Megaloblastic anemia C. Hyperparathyroidism D. Restless legs syndrome

Correct Answer: D. Restless legs syndrome

Question 18

Clinical Scenario:

A 67-year-old woman who was diagnosed with cancer 2 months ago presents to her oncologist with a 6-day history of numbness and tingling in her hands and feet. She is concerned that these symptoms may be related to progression of her cancer even though she has been faithfully following her chemotherapy regimen. She is not currently taking any other medications and has never previously experienced these symptoms. On physical exam, she is found to have decreased sensation to pinprick and fine touch over hands, wrists, ankles, and feet. Furthermore, she is found to have decreased reflexes throughout. Her oncologist assures her that these symptoms are a side effect from her chemotherapy regimen rather than progression of the cancer. The drug most likely responsible for her symptoms has which of the following mechanisms?

Answer Choices:

A. Alkylation of DNA B. DNA strand breaking C. Inhibit folate metabolism D. Inhibit microtubule formation

Correct Answer: D. Inhibit microtubule formation

Question 19

Clinical Scenario:

A 60-year-old woman presents with changes in her left breast that started 1 month ago. The patient states that she noticed that an area of her left breast felt thicker than before, and has not improved. She came to get it

checked out because her best friend was just diagnosed with invasive ductal carcinoma. The past medical history is significant for Hashimoto's thyroiditis, well-managed medically with levothyroxine. The patient has a 30-pack-year smoking history, but she quit over 15 years ago. The menarche occurred at age 11, and the menopause was at age 53. She does not have any children and has never been sexually active. Her last screening mammogram 10 months ago was normal. The family history is significant for her mother dying from a myocardial infarction (MI) at age 68, her sister dying from metastatic breast cancer at age 55, and for colon cancer in her paternal grandfather. The review of systems is notable for unintentional weight loss of 3.6 kg (8 lb) in the past month. The vital signs include: temperature 37.0°C (98.6°F), blood pressure 110/70 mm Hg, pulse 72/min, respiratory rate 15/min, and oxygen saturation 98% on room air. The physical examination is significant only for a minimally palpable mass with irregular, poorly defined margins in the upper outer quadrant of the left breast. The mass is rubbery and movable. There is no axillary lymphadenopathy noted. Which of the following characteristics is associated with this patient's most likely type of breast cancer in comparison to her friend's diagnosis?

Answer Choices:

A. Worse prognosis B. Can present bilaterally C. Higher prevalence D. Mammogram is more likely to demonstrate a discrete spiculated mass

Correct Answer: B. Can present bilaterally

Question 20

Clinical Scenario:

A 64-year-old woman otherwise healthy presents with acute onset severe rectal bleeding. The patient says that 2 hours ago bleeding began suddenly after a difficult bowel movement. She says the blood is bright red, and, initially, bleeding was brisk but now has stopped. The patient denies having any similar symptoms in the past. She has noticed that she bled more easily while having her regular manicure/pedicure for the past 3 months but thought it was nothing serious. No significant past medical history and the patient does not take any current medications. Family history is unremarkable. Review of systems is positive for mild dyspnea on exertion the past 2-3 months. Her vital signs include: temperature 37.0°C (98.6°F), blood pressure 100/65 mm Hg, pulse 95/min, respiratory rate 15/min, and oxygen saturation 97% on room air. A cardiac examination is significant for a 2/6 systolic murmur loudest at the right upper sternal border. Rectal exam shows no evidence of external hemorrhoids, fissures, or lesions. No active bleeding is noted. The stool is guaiac positive. Deficiency of which of the following is most likely the cause of this patient's condition?

Answer Choices:

A. Antithrombin III B. von Willebrand factor C. Factor VIII D. ADAMST13 gene mutation

Correct Answer: B. von Willebrand factor

Question 21

Clinical Scenario:

A 23-year-old woman presents to your office for a gynecological exam. She says that she has been in good health and has no complaints. She has been in a steady monogamous relationship for the past year. Physical examination was unremarkable. Screening tests are performed and return positive for gonorrhea. You treat her with an intramuscular injection of ceftriaxone and 7 day course of doxycycline. What else is recommended for this case?

Answer Choices:

A. Treatment with penicillin G for potential co-infection with syphilis B. Treat her partner for gonorrhea and chlamydia C. Inform her that her partner is likely cheating on her D. Perform an abdominal ultrasonography in order to rule out pelvic inflammatory disease

Correct Answer: B. Treat her partner for gonorrhea and chlamydia

Question 22

Clinical Scenario:

A 78-year-old right-handed man with hypertension and hyperlipidemia is brought to the emergency department for sudden onset of nausea and vertigo one hour ago. Physical examination shows 5/5 strength in all extremities. Sensation to light touch and pinprick is decreased in the right arm and leg. A CT scan of the brain shows an acute infarction in the distribution of the left posterior cerebral artery. Further evaluation of this patient is most likely to show which of the following findings?

Answer Choices:

A. Right-sided homonymous hemianopia B. Left-sided gaze deviation C. Prosopagnosia D. Right-sided superior quadrantanopia

Correct Answer: A. Right-sided homonymous hemianopia

Question 23

Clinical Scenario:

A 24-year-old man presents with low-grade fever and shortness of breath for the last 3 weeks. Past medical history is significant for severe mitral regurgitation status post mitral valve replacement five years ago. His temperature is 38.3°C (101.0°F) and respiratory rate is 18/min. Physical examination reveals vertical hemorrhages under his nails, multiple painless erythematous lesions on his palms, and two tender, raised nodules on his fingers. Cardiac auscultation reveals a new-onset 2/6 holosystolic murmur loudest at the apex with the patient in the left lateral decubitus position. A transesophageal echocardiogram reveals vegetations on the prosthetic valve. Blood cultures reveal catalase-positive, gram-positive cocci. Which of the following characteristics is associated with the organism most likely responsible for this patient's condition?

Answer Choices:

A. Hemolysis B. Coagulase positive C. DNase positive D. Novobiocin sensitive

Correct Answer: D. Novobiocin sensitive

Question 24

Clinical Scenario:

One day after delivery, an African American female newborn develops yellow discoloration of the eyes. She was born at term via uncomplicated vaginal delivery and weighed 3.4 kg (7 lb 8 oz). Her mother did not receive prenatal care. Examination shows scleral icterus and mild hepatosplenomegaly. Laboratory studies show: Hemoglobin 10.7 mg/dL Reticulocytes 3.5% Maternal blood group O, Rh-negative Anti-Rh antibody titer positive Fetal blood group A, Rh-negative Serum Bilirubin, total 6.1 mg/dL Direct 0.4 mg/dL Which of the following is the most likely cause of this patient's condition?"

Answer Choices:

A. Viral infiltration of the bone marrow B. Polymerization of deoxygenated hemoglobin C. Atresia of the biliary tract D. Transfer of Anti-A antibodies

Correct Answer: D. Transfer of Anti-A antibodies

Question 25

Clinical Scenario:

A 19-year-old woman comes to the physician because of episodic, bilateral finger pain and discoloration that occurs with cold weather. Her fingers first turn white, then blue, before eventually returning to a normal skin color. The symptoms have been occurring daily and limit her ability to work. She has no history of serious illness and takes no medication. She does not smoke. Physician examination shows normal capillary refill of the nail beds. The radial pulse is palpable bilaterally. The remainder of the examination shows no abnormalities. Which of the following is the most appropriate pharmacotherapy for this patient?

Answer Choices:

A. Phenylephrine B. Isosorbide dinitrate C. Nifedipine D. Ergotamine

Correct Answer: C. Nifedipine

Question 26

Clinical Scenario:

A 52-year-old woman is brought to the emergency department by her husband because of weakness, abdominal pain, and a productive cough for 4 days. She also reports increased urination for the past 2 days. This morning, she had nausea and five episodes of vomiting. She has type 1 diabetes mellitus and hypertension. Current medications include insulin and lisinopril. She admits to have forgotten to take her medication in the last few days. Her temperature is 38.4°C (101.1°F), pulse is 134/min, respirations 31/min, and blood pressure is 95/61 mm Hg. Examination shows dry mucous membranes and decreased skin turgor. Abdominal examination shows diffuse tenderness with no guarding or rebound. Bowel sounds are normal. Laboratory studies show: Serum

Na⁺ 139 mEq/L K⁺ 5.3 mEq/L Cl⁻ 106 mEq/L Glucose 420 mg/dL Creatinine 1.0 mg/dL Urine Blood negative Glucose 4+ Ketones 3+ Arterial blood gas analysis on room air shows: pH 7.12 pCO₂ 17 mm Hg pO₂ 86 mm Hg HCO₃⁻ 12 mEq/L Which of the following is the most likely underlying cause of this patient's increased potassium?"

Answer Choices:

A. Increased renal potassium absorption B. Muscle cell breakdown C. Extracellular potassium shift D. Repeated vomiting

Correct Answer: C. Extracellular potassium shift

Question 27

Clinical Scenario:

A 24-year-old woman comes to the physician for a routine health maintenance examination. She feels well. Menses occur at regular 28-day intervals and last for 3–5 days, with normal flow. They are occasionally accompanied by pain. Three years ago, she was diagnosed with chlamydial cervicitis and treated with doxycycline. She has been sexually active with multiple partners since the age of 18 years. She regularly uses condoms for contraception. She drinks 2–3 beers on weekends and smokes half a pack of cigarettes daily. Vital signs are within normal limits. Physical examination including a complete pelvic exam shows no abnormalities. A Pap smear shows a low-grade squamous epithelial lesion (LSIL). Which of the following is the most appropriate next step in management?

Answer Choices:

A. Colposcopy with endocervical sampling B. Colposcopy with endocervical and endometrial sampling C. Repeat Pap smear in 12 months D. Repeat Pap smear in 3 years

Correct Answer: C. Repeat Pap smear in 12 months

Question 28

Clinical Scenario:

A 4-year-old boy is brought to the pediatrician with fever, diarrhea and bilateral red eye for 7 days. His parents noted that he has never had an episode of diarrhea this prolonged, but several other children at daycare had been ill. His immunization history is up to date. His vitals are normal except for a temperature of 37.5°C (99°F). A physical exam is significant for mild dehydration, preauricular adenopathy, and bilateral conjunctival injection with watery discharge. What is the most likely diagnosis?

Answer Choices:

A. Rotavirus infection B. C. difficile colitis C. Adenovirus infection D. Vibrio parahaemolyticus infection

Correct Answer: C. Adenovirus infection

Question 29

Clinical Scenario:

A 55-year-old obese woman is referred to the cardiology clinic for progressive dyspnea. She has had no recent travel or sick contacts. Besides a multivitamin, she has only tried online weight-loss medications for the past five years, including fenfluramine-phentermine. An echocardiogram reveals a dilated right ventricle with systolic pressure of 60 mmHg as well as both tricuspid and pulmonary regurgitation. A right heart catheterization shows a mean pulmonary artery pressure of 40 mmHg. What disease process is most analogous to this patient's presentation?

Answer Choices:

A. Subacute endocarditis B. Carcinoid syndrome C. Left heart failure D. Chronic obstructive pulmonary disease

Correct Answer: B. Carcinoid syndrome

Question 30

Clinical Scenario:

A 45-year-old man with type 2 diabetes mellitus presents to his family physician for a follow-up appointment. He is currently using a 3-drug regimen consisting of metformin, sitagliptin, and glipizide. Despite this therapeutic regimen, his most recent hemoglobin A1c level is 8.1%. Which of the following is the next best step for this patient?

Answer Choices:

A. Discontinue glipizide; initiate insulin glargine 10 units at bedtime B. Discontinue metformin; initiate insulin aspart at mealtimes C. Discontinue sitagliptin; initiate basal-bolus insulin D. Discontinue metformin; initiate insulin glargine 10 units at bedtime

Correct Answer: A. Discontinue glipizide; initiate insulin glargine 10 units at bedtime

Question 31

Clinical Scenario:

A 57-year-old man presents to the emergency department after an episode of syncope. He states that he was at home when he suddenly felt weak and experienced back pain that has been persistent. He states that he vomited forcefully several times after the episode. The patient has a past medical history of diabetes, hypertension, dyslipidemia, and depression. He smokes 1.5 packs of cigarettes per day and drinks 10 alcoholic beverages each night. His temperature is 97.5°F (36.4°C), blood pressure is 107/48 mmHg, pulse is 130/min, respirations are 19/min, and oxygen saturation is 99% on room air. A chest radiograph is within normal limits. Physical exam is notable for abdominal tenderness and a man resting in an antalgic position. Urinalysis is currently pending but reveals a concentrated urine sample. Which of the following is the most likely diagnosis?

Answer Choices:

A. Abdominal aortic aneurysm B. Boerhaave syndrome C. Nephrolithiasis D. Pancreatitis

Correct Answer: A. Abdominal aortic aneurysm

Question 32**Clinical Scenario:**

A 3-year-old boy is brought for general developmental evaluation. According to his parents he is playing alongside other children but not in a cooperative manner. He has also recently begun to ride a tricycle. Upon questioning you also find that he is toilet trained and can stack 9 blocks. Upon examination you find that he can copy a circle though he cannot yet copy a triangle or draw stick figures. In addition he is currently speaking in two word phrases but cannot yet use simple sentences. Based on these findings you tell the parents that their child's development is consistent with which of the following?

Answer Choices:

A. Normal social, normal motor, normal language B. Normal social, delayed motor, delayed language C. Normal social, normal motor, delayed language D. Delayed social, normal motor, delayed language

Correct Answer: C. Normal social, normal motor, delayed language

Question 33**Clinical Scenario:**

A 14-year-old Asian girl is brought to the physician because of a 6-week history of fatigue. During this period, she has had a 3-kg (6.6-lb) weight loss and intermittent low-grade fevers. She also reports recurrent episodes of pain in her left wrist and right knee. She has no personal history of serious illness. Her aunt has rheumatoid arthritis. The patient appears pale. Her temperature is 38°C (100.4°F). Examination shows diffuse lymphadenopathy. Oral examination shows several painless oral ulcers. The left wrist and the right knee are swollen and tender to touch. The remainder of the examination shows no abnormalities. Laboratory studies show a hemoglobin concentration of 10 g/dL, a leukocyte count of 3,000/mm³, and a platelet count of 80,000/mm³. Urinalysis shows excessive protein. Further evaluation of this patient is most likely to show which of the following findings?

Answer Choices:

A. Anti-citrullinated peptide antibodies B. Positive monospot test C. Anti-dsDNA antibodies D. Elevated serum IgA levels

Correct Answer: C. Anti-dsDNA antibodies

Question 34

Clinical Scenario:

A 79-year-old male presents to your office for his annual flu shot. On physical exam you note several linear bruises on his back. Upon further questioning he denies abuse from his daughter and son-in-law, who live in the same house. The patient states he does not want this information shared with anyone. What is the most appropriate next step, paired with its justification?

Answer Choices:

A. Breach patient confidentiality, as this patient is a potential victim of elder abuse and that is always reportable
B. Do not break patient confidentiality, as this would potentially worsen the situation
C. Do not break patient confidentiality, as elder abuse reporting is not mandatory
D. See the patient back in 2 weeks and assess whether the patient's condition has improved, as his condition is not severe

Correct Answer: A. Breach patient confidentiality, as this patient is a potential victim of elder abuse and that is always reportable

Question 35

Clinical Scenario:

A 25-year-old man is brought to the emergency department by the police after a motor vehicle accident. He was reportedly speeding in a residential area and collided with a tree. He was later found by police naked in the street, screaming "shoot me so the devil will leave". A review of his medical record is unremarkable. At the hospital, he continues to act agitated and bizarre. His temperature is 37.0°C (98.6°F), the blood pressure is 140/86 mm Hg, and the heart rate is 90/min. The physical exam is notable for agitation, pacing around the room, occasionally yelling at the staff to help him "kill the devil". An ocular exam is significant for mild horizontal nystagmus. The patient appears to be drooling and has some difficulty with coordination. Which of the following is the most likely cause of this patient's presentation?

Answer Choices:

A. Cocaine intoxication
B. Central nervous system infection
C. Phencyclidine (PCP) intoxication
D. Serotonin syndrome

Correct Answer: C. Phencyclidine (PCP) intoxication

Question 36

Clinical Scenario:

A 61-year-old man comes to the physician because of a 2-month history of a cough productive of clear mucoid sputum. He has smoked one pack of cigarettes daily for 33 years. Physical examination shows no abnormalities. Chest x-ray shows a 2-cm solid nodule in the periphery of the lower left lobe. A bronchial biopsy of the mass shows numerous mucin-filled epithelial cells lining the alveolar basement membrane. The

cells have prominent nucleoli, coarse chromatin, and some cells have multiple nuclei. Which of the following is the most likely diagnosis?

Answer Choices:

A. Small cell carcinoma B. Pulmonary hamartoma C. Adenocarcinoma in situ D. Carcinoid tumor

Correct Answer: C. Adenocarcinoma in situ

Question 37

Clinical Scenario:

A 16-year-old woman is brought to the emergency department by her family for not being responsive. The patient had locked herself in her room for several hours after breaking up with her boyfriend. When her family found her, they were unable to arouse her and immediately took her to the hospital. The patient has a past medical history of anorexia nervosa, which is being treated, chronic pain, and depression. She is not currently taking any medications. The patient has a family history of depression in her mother and grandmother. IV fluids are started, and the patient seems to be less somnolent. Her temperature is 101°F (38.3°C), pulse is 112/min, blood pressure is 90/60 mmHg, respirations are 18/min, and oxygen saturation is 95% on room air. On physical exam, the patient is somnolent and has dilated pupils and demonstrates clonus. She has dry skin and an ultrasound of her bladder reveals 650 mL of urine. The patient is appropriately treated with sodium bicarbonate. Which of the following is the best indicator of the extent of this patient's toxicity?

Answer Choices:

A. Liver enzyme elevation B. QRS prolongation C. QT prolongation D. Serum drug level

Correct Answer: B. QRS prolongation

Question 38

Clinical Scenario:

A 55-year-old woman comes to the physician 10 days after noticing a mass in her left breast while bathing. She is concerned that it is breast cancer because her sister was diagnosed with breast cancer 3 years ago at 61 years of age. Menopause occurred 6 months ago. She has smoked 2 packs of cigarettes daily for 30 years. She took an oral contraceptive for 20 years. Current medications include hormone replacement therapy and a calcium supplement. Examination shows a 2.5-cm, palpable, hard, nontender, mass in the upper outer quadrant of the left breast; there is tethering of the skin over the lump. Examination of the right breast and axillae shows no abnormalities. Mammography shows an irregular mass with microcalcifications and oil cysts. A core biopsy shows foam cells and multinucleated giant cells. Which of the following is the most appropriate next step in management?

Answer Choices:

A. Neoadjuvant chemotherapy B. Reassurance C. Modified radical mastectomy D. Wide excision of the lump

Correct Answer: B. Reassurance

Question 39

Clinical Scenario:

During a humanitarian mission to southeast Asia, a 42-year-old man is brought to the outpatient clinic for a long history (greater than 2 years) of progressive, painless, enlargement of his scrotum. The family history is negative for malignancies and inheritable diseases. The personal history is relevant for cigarette smoking (up to 2 packs per day for the last 20 years) and several medical consultations for an episodic fever that resolved spontaneously. The physical examination is unremarkable, except for an enlarged left hemiscrotum that transilluminates. Which of the following accounts for the underlying mechanism in this patient's condition?

Answer Choices:

A. Autoimmune B. Invasive neoplasm C. Decreased lymphatic fluid absorption D. Patent processus vaginalis

Correct Answer: C. Decreased lymphatic fluid absorption

Question 40

Clinical Scenario:

A 64-year-old man presents to his primary care physician for a fall. The patient states that he has felt abnormally clumsy lately and has noticed himself tripping and bumping into things. He states he otherwise is healthy but admits to having unprotected sex with multiple people recently. His temperature is 99.5°F (37.5°C), blood pressure is 127/68 mm Hg, pulse is 100/min, respiratory rate is 24/min, and oxygen saturation is 98% on room air. Laboratory values are ordered as seen below.

Hemoglobin: 9 g/dL Hematocrit: 30% Mean corpuscular volume: 110 fL Leukocyte count: 6,500/mm³ with normal differential Platelet count: 197,000/mm³ AST: 15 U/L ALT: 22 U/L GGT: 10 U/L

Physical exam is notable for a broad-based and unstable gait. Which of the following conditions is the most likely etiology of this patient's presentation?

Answer Choices:

A. Chronic alcoholism B. Chronic gastritis C. Tertiary syphilis D. Vegetarian diet

Correct Answer: B. Chronic gastritis

Question 41

Clinical Scenario:

A 15-year-old African American boy presents to a pediatrician with complaints of yellow discoloration of the sclerae for the last 3 days. His mother informs the pediatrician that the boy developed prolonged jaundice during the neonatal period. On physical examination, vital signs are stable and general examination shows mild icterus and pallor. Examination of the abdomen suggests mild splenomegaly. Laboratory results are as follows:

Hemoglobin 9.9 g/dL Total leukocyte count 7,500/mm³ Platelet count 320,000/mm³ Reticulocyte count 5% Mean corpuscular hemoglobin 27.7 pg/cell Mean corpuscular hemoglobin concentration 32% g/dL Mean corpuscular volume 84 μm³ Serum total bilirubin 4.2 mg/dL Serum direct bilirubin 0.3 mg/dL Coombs test Negative Peripheral smear shows polychromasia, blister cells, and Heinz bodies. An abdominal ultrasonogram shows the presence of gallstones. Which of the following tests is most likely to be useful in diagnosing this patient?

Answer Choices:

A. Glycerol lysis test B. Methemoglobin reduction test C. Serum thyroxine, triiodothyronine, and thyroid-stimulating hormone D. Hepatoiminodiacetic acid scanning

Correct Answer: B. Methemoglobin reduction test

Question 42

Clinical Scenario:

A 36-year-old woman, gravida 3, para 2, at 37 weeks' gestation comes to the emergency department because of sparse vaginal bleeding for 3 hours. She also noticed the bleeding 3 days ago. She has had no prenatal care. Both of her previous children were delivered by lower segment transverse cesarean section. Her temperature is 37.1°C (98.8°F), pulse is 90/min, respirations are 16/min, and blood pressure is 110/80 mm Hg. The abdomen is nontender, and no contractions are felt. Examination shows that the fetus is in a vertex presentation. The fetal heart rate is 160/min and shows no abnormalities. Which of the following is the most appropriate next step in management?

Answer Choices:

A. Perform pelvic examination B. Perform transvaginal sonography C. Perform Kleihauer-Betke test D. Conduct contraction stress test

Correct Answer: B. Perform transvaginal sonography

Question 43

Clinical Scenario:

A 55-year-old man with type 2 diabetes mellitus comes to the physician because of a 4-day history of fever, chills, nausea, and abdominal pain. He does not use illicit drugs. His temperature is 39°C (102.2°F). Physical examination shows right upper quadrant tenderness. Ultrasonography of the abdomen shows a 6-cm solitary, fluid-filled cavity in the right hepatic lobe. CT-guided percutaneous aspiration of the cavity produces yellowish-green fluid. Culture of the aspirated fluid grows gram-negative, lactose-fermenting rods. Which of the following is the most likely cause of the color of the aspirated fluid?

Answer Choices:

A. Prodigiosin B. Myeloperoxidase C. Staphyloxanthin D. Biliverdin

Correct Answer: B. Myeloperoxidase

Question 44

Clinical Scenario:

A 68-year-old woman comes to the physician for a follow-up examination. Three months ago, she underwent heart transplantation for restrictive cardiomyopathy and was started on transplant rejection prophylaxis. Her pulse is 76/min and blood pressure is 148/82 mm Hg. Physical examination shows enlargement of the gum tissue. There is a well-healed scar on her chest. Serum studies show hyperlipidemia. The physician recommends removing a drug that decreases T cell activation by inhibiting the transcription of interleukin-2 from the patient's treatment regimen and replacing it with a different medication. Which of the following drugs is the most likely cause of the adverse effects seen in this patient?

Answer Choices:

A. Prednisolone B. Tacrolimus C. Cyclosporine D. Mycophenolate mofetil

Correct Answer: C. Cyclosporine

Question 45

Clinical Scenario:

A 39-year-old man comes to the physician for a follow-up examination. He was diagnosed with latent tuberculosis infection 3 months ago. He has had generalized fatigue and dyspnea on exertion for the past 6 weeks. He does not smoke and drinks 2–3 beers on weekends. Vital signs are within normal limits. Examination shows conjunctival pallor. Laboratory studies show: Hemoglobin 7.8 g/dL Mean corpuscular volume 72 μm^3 Red cell distribution width 17% (N = 13–15) Reticulocyte count 0.7% Leukocyte count 6,800/mm³ Platelet count 175,000/mm³ Serum Creatinine 0.8 mg/dL Iron 246 $\mu\text{g/dL}$ Ferritin 446 ng/mL Total iron-binding capacity 212 $\mu\text{g/dL}$ (N = 250–450) Which of the following is the most likely cause of this patient's symptoms?"

Answer Choices:

A. Iron deficiency B. Chronic inflammation C. Beta thalassemia minor D. Adverse effect of medication

Correct Answer: D. Adverse effect of medication

Question 46

Clinical Scenario:

A 22-year-old female college student comes to your clinic to establish care. She has no significant past medical history and her only complaint today is that she has had trouble maintaining a consistent weight. Her temperature is 98.6°F (37.0°C), blood pressure is 100/65 mmHg, pulse is 62/min, and respirations are 12/min. Her body mass index is 19.5. Her physical exam is significant for callused knuckles and dental enamel erosions. What laboratory abnormalities are likely to be found in this patient?

Answer Choices:

A. Decreased chloride, decreased potassium, decreased bicarbonate B. Decreased chloride, decreased potassium, increased bicarbonate C. Decreased chloride, increased potassium, increased bicarbonate D. Increased chloride, decreased potassium, decreased bicarbonate

Correct Answer: B. Decreased chloride, decreased potassium, increased bicarbonate

Question 47**Clinical Scenario:**

A 38-year-old woman comes to the physician because of a 4-month history of crampy abdominal pain, recurrent watery diarrhea, and a 2.5-kg (5.5-lb) weight loss. Her husband has noticed that after meals, her face and neck sometimes become red, and she develops shortness of breath and starts wheezing. Examination shows a grade 3/6 systolic murmur heard best at the left lower sternal border. The abdomen is soft, and there is mild tenderness to palpation with no guarding or rebound. Without treatment, this patient is at greatest risk of developing which of the following conditions?

Answer Choices:

A. Laryngeal edema B. Pigmented dermatitis C. Megaloblastic anemia D. T-cell lymphoma

Correct Answer: B. Pigmented dermatitis

Question 48**Clinical Scenario:**

A 66-year-old woman presents to the primary care physician with complaints of involuntary loss of urine. This has been occurring over the past month for no apparent reason while suddenly feeling the need to urinate. History reveals triggers that stimulate the desire to pass urine, such as running water, handwashing, and cold weather. There is no family history of similar symptoms in her mother or any of her 8 children. Her blood pressure is 130/80 mm Hg, heart rate is 72/min, respiratory rate is 22/min, and temperature is 36.6°C (98.0°F). Physical examination is unremarkable. Urinalysis reveals the following: Color Yellow Clarity/turbidity Clear pH 5.5 Specific gravity 1.015 Nitrites Negative Leukocyte esterase Negative Which of the following is the best next step in the management of this patient?

Answer Choices:

A. Administer antimuscarinics B. Bladder training C. Posterior tibial nerve stimulation D. Surgery

Correct Answer: B. Bladder training

Question 49**Clinical Scenario:**

A 75-year-old woman presents with episodic abdominal pain following meals for the past few years. She says these episodes have worsened over the past month. Past medical history is significant for type 2 diabetes mellitus diagnosed 30 years ago, managed with metformin. Her most recent HbA1C last month was 10%. Vital signs include: blood pressure 110/70 mm Hg, pulse 80/min, and respiratory rate 16/min. Physical examination is unremarkable. Which of the following is the most likely diagnosis in this patient?

Answer Choices:

A. Acute pancreatitis B. Hepatic infarction C. Chronic renal failure D. Mesenteric artery occlusion

Correct Answer: D. Mesenteric artery occlusion

Question 50

Clinical Scenario:

A 52-year-old woman comes to the physician because of abdominal discomfort, anorexia, and mild fatigue. She has systemic lupus erythematosus and takes hydroxychloroquine. She does not drink alcohol or use illicit drugs. Physical examination shows no abnormalities. Laboratory studies show: Alanine aminotransferase 455 U/L Aspartate aminotransferase 205 U/L Hepatitis B surface antigen positive Hepatitis B surface antibody negative Hepatitis B envelope antigen positive Hepatitis B core antigen IgG antibody positive Which of the following is the most appropriate pharmacotherapy for this patient?"

Answer Choices:

A. Pegylated interferon-gamma B. Acyclovir C. Tenofovir D. Sofosbuvir "

Correct Answer: C. Tenofovir

Question 51

Clinical Scenario:

A 58-year-old woman is brought to the emergency department 1 hour after she accidentally spilled hot oil on her leg while cooking. The Venezuelan receptionist reports that the patient only speaks and understands Spanish. She is accompanied by her adult son, who speaks English and Spanish. Her vital signs are within normal limits. Physical examination shows a 10 × 12-cm, erythematous, swollen patch of skin with ruptured blisters on the anterior aspect of the left leg. The physician considers administration of tetanus prophylaxis and wound debridement but cannot speak Spanish. Which of the following is the most appropriate action by the physician?

Answer Choices:

A. Wait for a licensed Spanish interpreter to communicate the treatment plan B. Communicate the treatment plan through the son C. Perform the treatment without prior communication D. Communicate the treatment plan through the receptionist

Correct Answer: A. Wait for a licensed Spanish interpreter to communicate the treatment plan

Question 52

Clinical Scenario:

A healthy 48-year-old presents for a well-patient visit. He has no symptoms and feels well. Past medical history is significant for asthma, chronic sinusitis, and nasal polyps. He occasionally takes diphenhydramine for allergies. Both of his parents and an elder brother are in good health. Today, his blood pressure is 119/81 mm Hg, heart rate is 101/min, respiratory rate is 21/min, and temperature 37°C (98.6°F). Routine screening blood work reveals elevated total cholesterol. The patient asks if he should take low-dose aspirin to reduce his risk of stroke and heart attack. Of the following, which is the best response?

Answer Choices:

A. Yes, aspirin therapy is recommended. B. Yes, but only every other day. C. No, because all chronic sinusitis carries aspirin-complications. D. Have you had a reaction to aspirin in the past?

Correct Answer: D. Have you had a reaction to aspirin in the past?

Question 53

Clinical Scenario:

A 48-year-old female complains of tingling sensation in her fingertips as well as the skin around her mouth which woke her up from sleep. She is in the postoperative floor as she just underwent a complete thyroidectomy for papillary thyroid cancer. Her temperature is 37° C (98.6° F), respirations are 15/min, pulse is 67/min, and blood pressure is 122/88 mm Hg. While recording the blood pressure, spasm of the muscles of the hand and forearm is seen. What is the next best step in the management of this patient?

Answer Choices:

A. Propylthiouracil B. Magnesium replacement C. Albumin infusion D. Calcium replacement

Correct Answer: D. Calcium replacement

Question 54

Clinical Scenario:

A 35-year-old man is brought into the emergency department by emergency medical services with his right hand wrapped in bloody bandages. The patient states that he is a carpenter and was cutting some wood for a home renovation project when he looked away and injured one of his digits with a circular table saw. He states that his index finger was sliced off and is being brought in by his wife. On exam, his vitals are within normal limits and stable, and he is missing part of his second digit on his right hand distal to the proximal interphalangeal joint. How should the digit be transported to the hospital for the best outcome?

Answer Choices:

A. Wrapped in a towel B. In a sterile bag of tap water C. In a sterile plastic bag wrapped in saline moistened gauze D. In a sterile plastic bag wrapped in saline moistened gauze on ice

Correct Answer: D. In a sterile plastic bag wrapped in saline moistened gauze on ice

Question 55

Clinical Scenario:

A 63-year-old man with a history of hypertension and atrial fibrillation is brought into the emergency room and found to have a ventricular tachyarrhythmia. Ibutilide is discontinued and the patient is switched to another drug that also prolongs the QT interval but is associated with a decreased risk of torsades de pointes. Which drug was most likely administered in this patient?

Answer Choices:

A. Digoxin B. Esmolol C. Amiodarone D. Quinidine

Correct Answer: C. Amiodarone

Question 56

Clinical Scenario:

A 64-year-old man is admitted with a history of altered mental status. He was in his usual state of health until a few days ago when he has started to become confused, lethargic, forgetful, and repeating the same questions. Over the last few days, he sometimes appears perfectly normal, and, at other times, he has difficulty recognizing his family members. Yesterday, he was screaming that the room was filled with snakes. Past medical history is significant for type 2 diabetes mellitus, managed medically, and chronic kidney disease, for which he undergoes regular hemodialysis on alternate days. There is no history of smoking, alcohol use, or illicit drug use. His vitals include: blood pressure 129/88 mm Hg, pulse 112/min, temperature 38.2°C (100.8°F), and respiratory rate 20/min. The patient is oriented only to person and place. His mini-mental state examination (MMSE) score is 18/30, where he had difficulty performing basic arithmetic calculations and recalled only 1 out of 3 objects. Nuchal rigidity is absent. Muscle strength is 5/5 bilaterally. Which of the following is the most likely diagnosis in this patient?

Answer Choices:

A. Delirium B. Dementia C. Transient global amnesia D. Wernicke's aphasia

Correct Answer: A. Delirium

Question 57

Clinical Scenario:

A 33-year-old man comes to the physician because of right scrotal swelling for the past 2 weeks. He has had mild lower abdominal discomfort for the past 3 weeks. There is no personal or family history of serious illness. He appears healthy. Vital signs are within normal limits. Examination shows gynecomastia. There is no

inguinal lymphadenopathy. There is a firm nontender nodule over the right testicle. When a light is held behind the scrotum, it does not shine through. When the patient is asked to cough, the nodule does not cause a bulge. The abdomen is soft and nontender. The liver is palpated 2 cm below the right costal margin. Digital rectal examination is unremarkable. Serum alpha-fetoprotein, LDH, and hCG levels are markedly elevated. An x-ray of the chest shows no abnormalities. Ultrasound of the testis shows a cystic 3-cm mass with variable echogenicity. A CT of the abdomen shows multiple hypoattenuating lesions on the liver and retroperitoneal lymph nodes. A radical inguinal orchiectomy with retroperitoneal lymph node dissection is performed. Which of the following is the most appropriate next step in management?

Answer Choices:

A. Radiation therapy B. Cisplatin, etoposide, and bleomycin therapy C. Leucovorin, 5-fluorouracil and oxaliplatin therapy D. Stem cell transplant

Correct Answer: B. Cisplatin, etoposide, and bleomycin therapy

Question 58

Clinical Scenario:

A 12-year-old boy presents to the pediatrician for a routine checkup. He and his family immigrated from Pakistan to the United States when he was 9 years of age. Per his mother, he had measles when he was 4 years of age and a high fever following a sore throat at the age 7. He received all appropriate vaccinations when he arrived in the United States. He takes no medications. He does well academically and plays soccer in a recreational league. He was born at 38 weeks gestation. His temperature is 98.4°F (36.9°C), blood pressure is 115/65 mmHg, pulse is 80/min, and respirations are 18/min. On exam, he is a healthy boy in no apparent distress. Breath sounds are equal bilaterally with good aeration. Fixed splitting of the second heart sound is noted on auscultation. Without adequate treatment, this patient will be at increased risk for developing which of the following?

Answer Choices:

A. Acute endocarditis B. Extra-cardiac left-to-right shunting C. Mitral stenosis D. Reversal of left-to-right shunting

Correct Answer: D. Reversal of left-to-right shunting

Question 59

Clinical Scenario:

A 22-year-old nulligravid woman comes to the physician for evaluation of irregular periods. Menarche was at the age of 12 years. Her menses have always occurred at variable intervals, and she has spotting between her periods. Her last menstrual period was 6 months ago. She has diabetes mellitus type 2 and depression. She is not sexually active. She drinks 3 alcoholic drinks on weekends and does not smoke. She takes metformin and sertraline. She appears well. Her temperature is 37°C (98.6°F), pulse is 82/min, respirations are 15/min, and blood pressure is 118/75 mm Hg. BMI is 31.5 kg/m². Physical exam shows severe cystic acne on her face and

back. There are dark, velvet-like patches on the armpits and neck. Pelvic examination is normal. A urine pregnancy test is negative. Which of the following would help determine the cause of this patient's menstrual irregularities?

Answer Choices:

A. Measurement of follicle-stimulating hormone B. Progesterone withdrawal test C. Measurement of thyroid-stimulating hormone D. Measurement of prolactin levels

Correct Answer: B. Progesterone withdrawal test

Question 60

Clinical Scenario:

A 72-year-old man is taken to the emergency room after losing consciousness. According to his wife, he suddenly complained of fluttering in his chest, lightheadedness, and profuse sweating while walking to the grocery store. He then turned gray, lost consciousness, and collapsed onto the ground. His medical history is significant for a prior anterior wall myocardial infarction 2 years ago that was complicated by severe left ventricular systolic dysfunction. His blood pressure is 80/50 mm Hg, the temperature is 36.7°C (98.0°F), and the carotid pulse is not palpable. An ECG was obtained and the results are shown in the picture.

Cardiopulmonary resuscitation is initiated and the patient is cardioverted to sinus rhythm with an external defibrillator. The patient regains consciousness and states there was no antecedent chest discomfort. Cardiac enzymes are negative and serum electrolytes are normal. Which of the following is the best next step for this patient?

Answer Choices:

A. Intravenous magnesium sulphate B. Implantable cardioverter-defibrillator C. Intravenous adenosine D. Temporary or permanent cardiac pacing

Correct Answer: B. Implantable cardioverter-defibrillator

Question 61

Clinical Scenario:

An 83-year-old bedbound man presents with a shallow open ulcer over his sacrum, with a red wound bed. Upon further examination, he also has areas of non-blanching redness on his lateral malleoli. Which of the following interventions would most likely have prevented his condition?

Answer Choices:

A. Nutritional supplementation B. Topical antibiotics C. Anti-coagulants D. Frequent repositioning

Correct Answer: D. Frequent repositioning

Question 62

Clinical Scenario:

A 58-year-old woman with a history of nephrolithiasis presents with fever and acute-onset right flank pain. The patient says that 2 days ago she developed sudden-onset right flank pain and nausea which has progressively worsened. She describes the pain as severe, colicky, localized to the right flank, and radiating to the groin. This morning she woke with a fever and foul-smelling urine. She has no significant past medical history. Vital signs are temperature 40.0°C (104.0°F), blood pressure 110/70 mm Hg, pulse 92/min, and respiratory rate 21/min. Physical examination shows severe right costovertebral angle tenderness. Her laboratory findings are significant for the following: WBC 12,500/mm³ RBC 4.20 x 10⁶/mm³ Hematocrit 41.5% Hemoglobin 14.0 g/dL Platelet count 225,000/mm³ Urinalysis: Color Dark yellow Clarity Clarity Turbid pH 5.9 Specific gravity 1.026 Glucose None Ketones None Nitrites Positive Leukocyte esterase Positive Bilirubin Negative Urobilinogen 0.6 mg/dL Protein Trace RBC 325/hpf WBC 8,200/hpf Bacteria Many A non-contrast CT of the abdomen and pelvis shows an obstructing 7-mm diameter stone lodged at the ureteropelvic junction. There is also evidence of hydronephrosis of the right kidney. Which of the following is the best course of treatment for this patient?

Answer Choices:

A. Discharge home with oral antibiotics B. Admit to hospital for IV antibiotics C. Administer potassium citrate D. Admit to hospital for percutaneous nephrostomy and IV antibiotics

Correct Answer: D. Admit to hospital for percutaneous nephrostomy and IV antibiotics

Question 63

Clinical Scenario:

A 2-week-old boy has developed bilious vomiting. He was born via cesarean section at term. On physical exam, his pulse is 140, blood pressure is 80/50 mmHg, and respirations are 40/min. His abdomen appears distended and appears diffusely tender to palpation. Abdominal imaging is obtained (Figures A). Which of the following describes the mechanism that caused this child's disorder?

Answer Choices:

A. Ischemia-reperfusion injury in premature neonate B. Telescoping segment of bowel C. Abnormal rotation of the midgut D. Partial absence of ganglion cells in large intestine

Correct Answer: C. Abnormal rotation of the midgut

Question 64

Clinical Scenario:

A research group is investigating an allosteric modulator to improve exercise resistance and tolerance at low-oxygen conditions. The group has created cultures of myocytes derived from high-performance college

athletes. The application of this compound to these cultures in a low-oxygen environment and during vigorous contraction leads to longer utilization of glucose before reaching a plateau and cell death; however, the culture medium is significantly acidified in this experiment. An activating effect on which of the following enzymes would explain these results?

Answer Choices:

A. Pyruvate dehydrogenase B. Bisphosphoglycerate mutase C. Malate dehydrogenase D. Lactate dehydrogenase

Correct Answer: D. Lactate dehydrogenase

Question 65

Clinical Scenario:

A 32-year-old man who recently emigrated from Colombia comes to the physician because of a 3-month history of shortness of breath and fatigue. Physical examination shows jugular venous distention and an additional late diastolic heart sound. Crackles are heard at the lung bases bilaterally. Cardiac catheterization is performed and left ventricular pressures are obtained. The left ventricular pressure-volume relationship compared to that of a healthy patient is shown. Which of the following is the most likely cause of this patient's heart failure?

Answer Choices:

A. Chagas heart disease B. Viral myocarditis C. Cardiac sarcoidosis D. Thiamine deficiency

Correct Answer: C. Cardiac sarcoidosis

Question 66

Clinical Scenario:

A 50-year-old man presents to his primary care physician with a chief complaint of chest pain that is squeezing in nature. He used to have similar symptoms in the past while playing tennis with his friends. Yesterday, while moving furniture in his new home, he experienced this pain that lasted for 20 minutes and radiated towards his jaw and shoulder. He has been diagnosed with diabetes mellitus and hypertension for over 10 years and regularly takes his medications. The pain is not associated with nausea, vomiting, food intake, sweating, or cough. On physical examination, the patient is not in acute distress. His blood pressure is 135/85 mm Hg, heart rate is 80/min, respiratory rate is 16/min, temperature is 36.9°C (98.5°F), and BMI is 30 kg/m². On physical examination, bilateral vesicular breath sounds are heard with absent chest tenderness. Cardiovascular examination reveals normal S1 and S2 without any abnormal sounds or murmur. Abdominal examination is within normal limit. What is the most likely cause of this patient's condition?

Answer Choices:

A. GERD B. Musculoskeletal pain C. Anxiety D. Myocardial ischemia

Correct Answer: D. Myocardial ischemia

Question 67

Clinical Scenario:

A 27-year-old man presents to the emergency department with severe substernal pain at rest, which radiates to his left arm and jaw. He reports that he has had similar but milder pain several times in the past during strenuous exercise. He had heart transplantation due to dilatory cardiomyopathy 5 years ago with an acute rejection reaction that was successfully treated with corticosteroids. He had been taking 1 mg tacrolimus twice a day for 3.5 years but then discontinued it and had no regular follow-ups. The man does not have a family history of premature coronary artery disease. His blood pressure is 110/60 mm Hg, heart rate is 97/min, respiratory rate is 22/min, and temperature is 37.3°C (99.1°F). On physical examination, the patient is alert, responsive, and agitated. Cardiac auscultation reveals a fourth heart sound (S4) and an irregularly irregular heart rhythm. His ECG shows ST elevation in leads I, II, V5, and V6, and ST depression in leads III and aVF. His complete blood count and lipidogram are within normal limits. The patient's cardiac troponin I and T levels are elevated. A coronary angiogram reveals diffuse concentric narrowing of all branches of the left coronary artery. What is the most likely causative mechanism of this patient's cardiac ischemia?

Answer Choices:

- A. Vasospasm of distal coronary arteries branches
- B. Left ventricular hypertrophy
- C. Obliterative arteriopathy
- D. Increased oxygen demand due to tachycardia

Correct Answer: C. Obliterative arteriopathy

Question 68

Clinical Scenario:

A 17-year-old female presents to your office expressing concern that despite experiencing monthly pelvic pain for the past few years, she has not yet started her menstrual cycle. She is not taking oral contraceptive therapy and has never been sexually active. On physical exam the patient is of normal stature with appropriate breast development and growth of pubic and underarm hair. The patient declined a vaginal exam. Karyotype analysis reveals she has 46 XX. Pregnancy test is negative, thyroid stimulating hormone, prolactin, luteinizing hormone (LH) and follicle-stimulating hormone (FSH) levels are normal. The uterus is normal on ultrasound. What is the likely cause of this patient's primary amenorrhea?

Answer Choices:

- A. Failure in development of Mullerian duct
- B. Failed canalization of external vaginal membrane
- C. Androgen insensitivity
- D. Pituitary infarct

Correct Answer: B. Failed canalization of external vaginal membrane

Question 69

Clinical Scenario:

A previously healthy 61-year-old man comes to the physician because of a 3-month history of intermittent fever, easy fatigability, and a 4.4-kg (9.7-lb) weight loss. Physical examination shows conjunctival pallor. The spleen is palpated 5 cm below the left costal margin. Laboratory studies show a leukocyte count of 75,300/mm³ with increased basophils, a platelet count of 455,000/mm³, and a decreased leukocyte alkaline phosphatase score. A peripheral blood smear shows increased numbers of promyelocytes, myelocytes, and metamyelocytes. Which of the following is the most likely diagnosis?

Answer Choices:

A. Chronic lymphocytic leukemia B. Essential thrombocythemia C. Chronic myeloid leukemia D. Acute promyelocytic leukemia

Correct Answer: C. Chronic myeloid leukemia

Question 70

Clinical Scenario:

A 41-year-old woman comes to the primary care physician's office with a 7-day history of headaches, sore throat, diarrhea, fatigue, and low-grade fevers. The patient denies any significant past medical history, recent travel, or recent sick contacts. On review of systems, the patient endorses performing sex acts in exchange for money and recreational drugs over the last several months. You suspect primary HIV infection, but the patient refuses further evaluation. At a follow-up appointment 1 week later, she reports that she had been previously tested for HIV, and it was negative. Physical examination does not reveal any external abnormalities of her genitalia. Her heart and lung sounds are normal on auscultation. Her vital signs show a blood pressure of 123/82 mm Hg, heart rate of 82/min, and a respiratory rate of 16/min. Of the following options, which is the next best step in patient management?

Answer Choices:

A. Repeat rapid HIV at this office check-up B. Retest with ELISA and Western blot in 2.5–8.5 weeks and again in 6 months C. Perform monospot test D. Perform VDRL

Correct Answer: B. Retest with ELISA and Western blot in 2.5–8.5 weeks and again in 6 months

Question 71

Clinical Scenario:

A 34-year-old woman comes to the physician because of a 3-month history of fatigue and a 4.5-kg (10-lb) weight loss despite eating more than usual. Her pulse is 115/min and blood pressure is 140/60 mm Hg. Physical examination shows warm, moist skin, and a diffuse, non-tender swelling over the anterior neck. Ophthalmologic examination shows swelling of the eyelids and proptosis bilaterally. Which of the following is the most likely cause of this patient's symptoms?

Answer Choices:

A. Nongranulomatous thyroid inflammation B. Thyrotropin receptor autoantibodies C. Parafollicular cell hyperplasia D. Thyroid peroxidase autoantibodies

Correct Answer: B. Thyrotropin receptor autoantibodies

Question 72

Clinical Scenario:

A physician scientist is looking for a more efficient way to treat HIV. Patients infected with HIV mount a humoral immune response by producing antibodies against the HIV envelope proteins. These antibodies are the same antibodies detected by the ELISA and western blot assays used to diagnose the disease. The physician scientist is trying to generate a new, more potent antibody against the same HIV envelope proteins targeted by the natural humoral immune response. Of the following proteins, which is the most likely target of the antibody he is designing?

Answer Choices:

A. gp120 B. CXCR4 C. p24 D. p17

Correct Answer: A. gp120

Question 73

Clinical Scenario:

An obese 52-year-old man is brought to the emergency department because of increasing shortness of breath for the past 8 hours. Two months ago, he noticed a mass on the right side of his neck and was diagnosed with laryngeal cancer. He has smoked two packs of cigarettes daily for 27 years. He drinks two pints of rum daily. He appears ill. He is oriented to person, place, and time. His temperature is 37°C (98.6°F), pulse is 111/min, respirations are 34/min, and blood pressure is 140/90 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 89%. Examination shows a 9-cm, tender, firm subglottic mass on the right side of the neck. Cervical lymphadenopathy is present. His breathing is labored and he has audible inspiratory stridor but is able to answer questions. The lungs are clear to auscultation. Arterial blood gas analysis on room air shows: pH 7.36 PCO₂ 45 mm Hg PO₂ 74 mm Hg HCO₃⁻ 25 mEq/L He has no advanced directive. Which of the following is the most appropriate next step in management?"

Answer Choices:

A. Tracheal stenting B. Tracheostomy C. Intramuscular epinephrine D. Cricothyroidotomy

Correct Answer: B. Tracheostomy

Question 74

Clinical Scenario:

A 45-year-old woman is in a high-speed motor vehicle accident and suffers multiple injuries to her extremities and abdomen. In the field, she was bleeding profusely and, upon arrival to the emergency department, she is lethargic and unable to speak. Her blood pressure on presentation is 70/40 mmHg. The trauma surgery team recommends emergency exploratory laparotomy. While the patient is in the trauma bay, her husband calls and says that the patient is a Jehovah's witness and that her religion does not permit her to receive a blood transfusion. No advanced directives are available. Which of the following is an appropriate next step?

Answer Choices:

A. Provide transfusions as needed B. Withhold transfusion based on husband's request C. Obtain an ethics consult D. Obtain a court order for transfusion

Correct Answer: A. Provide transfusions as needed

Question 75

Clinical Scenario:

A 39-year-old man presents to his primary care physician with a 10-hour history of severe diarrhea. He says that he was recently at a company picnic and after returning home he began to experience severe watery diarrhea. He says that the diarrhea was accompanied by nausea and abdominal pain. His physician informs him that he was likely infected by a lactose-fermenting, gram-negative organism. Which of the following changes would be seen in a cell that was affected by the heat stable toxin produced by this organism?

Answer Choices:

A. Decreased cyclic adenosine monophosphate B. Increased calcium C. Increased cyclic adenosine monophosphate D. Increased cyclic guanosine monophosphate

Correct Answer: D. Increased cyclic guanosine monophosphate

Question 76

Clinical Scenario:

A 2-week-old boy presents to the pediatrics clinic. The medical records notes a full-term delivery, however, the boy was born with chorioretinitis and swelling and calcifications in his brain secondary to an in utero infection. A drug exists that can be used to prevent infection by the pathogen responsible for this neonate's findings. This drug can also provide protection against infection by what other microorganism?

Answer Choices:

A. Mycobacterium tuberculosis B. Mycobacterium avium complex C. Pneumocystis jiroveci D. Cytomegalovirus

Correct Answer: C. Pneumocystis jiroveci

Question 77

Clinical Scenario:

A 63-year-old man with non-Hodgkin lymphoma is brought to the emergency department because of fever and confusion that have progressively worsened over the past 3 days. He also has a 3-day history of loose stools. He returned from France 2 weeks ago where he stayed in the countryside and ate typical French cuisine, including frog, snail, and various homemade cheeses. His last chemotherapy cycle was 3 weeks ago. He is oriented to person but not to place or time. His temperature is 39.5°C (103.1°F), pulse is 110/min, and blood pressure is 100/60 mm Hg. Examination shows cervical and axillary lymphadenopathy. The lungs are clear to auscultation. There is involuntary flexion of the bilateral hips and knees with passive flexion of the neck. Neurologic examination shows no focal findings. Laboratory studies show: Hemoglobin 9.3 g/dL Leukocyte count 3600/mm³ Platelet count 151,000/mm³ Serum Na⁺ 134 mEq/L Cl⁻ 103 mEq/L K⁺ 3.7 mEq/L Glucose 102 mg/dL Creatinine 1.3 mg/dL A lumbar puncture is performed. Cerebrospinal fluid analysis shows a leukocyte count of 1200/mm³ (76% segmented neutrophils, 24% lymphocytes), a protein concentration of 113 mg/dL, and a glucose concentration of 21 mg/dL. The results of blood cultures are pending. Which of the following is the most appropriate initial pharmacotherapy?"

Answer Choices:

A. Ampicillin and cefotaxime B. Acyclovir and dexamethasone C. Acyclovir D. Vancomycin, ampicillin, and cefepime

Correct Answer: D. Vancomycin, ampicillin, and cefepime

Question 78

Clinical Scenario:

A 28-year-old woman comes to the physician because of a two-month history of fatigue and low-grade fevers. Over the past 4 weeks, she has had increasing shortness of breath, a productive cough, and a 5.4-kg (11.9-lb) weight loss. Three months ago, the patient returned from a two-month trip to China. The patient appears thin. Her temperature is 37.9°C (100.2°F), pulse is 75/min, and blood pressure is 125/70 mm Hg. Examination shows lymphadenopathy of the anterior and posterior cervical chain. Rales are heard at the left lower lobe of the lung on auscultation. Laboratory studies show a leukocyte count of 11,300/mm³ and an erythrocyte sedimentation rate of 90 mm/h. An x-ray of the chest shows a patchy infiltrate in the left lower lobe and ipsilateral hilar enlargement. Microscopic examination of the sputum reveals acid-fast bacilli; polymerase chain reaction is positive. Sputum cultures are pending. After placing the patient in an airborne infection isolation room, which of the following is the most appropriate next step in management?

Answer Choices:

A. Await culture results before initiating treatment B. Perform interferon-γ release assay C. Obtain CT scan of the chest D. Administer isoniazid, rifampin, pyrazinamide, and ethambutol for 2 months, followed by isoniazid and rifampin for 4 months

Correct Answer: D. Administer isoniazid, rifampin, pyrazinamide, and ethambutol for 2 months, followed by isoniazid and rifampin for 4 months

Question 79

Clinical Scenario:

A hospitalized 45-year-old man has had mild flank pain since awakening 3 hours ago. He also reports a new generalized rash. Two weeks ago, he was diagnosed with pulmonary tuberculosis. Current medications include isoniazid, pyrazinamide, rifampin, ethambutol, and pyridoxine. His temperature is 38.3°C (100.9°F), pulse is 74/min, and blood pressure is 128/72 mm Hg. Examination of the skin shows diffuse erythema with confluent papules. There is no costovertebral angle tenderness. Laboratory studies show: Leukocyte count 9,800/mm³ Segmented neutrophils 59% Bands 3% Eosinophils 4% Lymphocytes 29% Monocytes 5% Serum Urea nitrogen 25 mg/dL Creatinine 1.9 mg/dL Urine WBC 8–10/hpf Eosinophils numerous RBC 5–6/hpf RBC casts negative WBC casts numerous In addition to intravenous fluid resuscitation, which of the following is the most appropriate next step in management?"

Answer Choices:

A. Initiate hemodialysis B. Administer ciprofloxacin C. Discontinue rifampin D. Perform renal biopsy

Correct Answer: C. Discontinue rifampin

Question 80

Clinical Scenario:

A 55-year-old man with atrial fibrillation is brought to the emergency department by his wife 6 hours after the acute onset of right arm weakness and slurred speech. An MRI of the brain shows a thrombus in the left middle cerebral artery. Twelve hours later, the patient develops ventricular tachycardia. Despite appropriate care, he dies. Which of the following histopathologic changes are most likely to be seen on a biopsy specimen from the affected brain tissue?

Answer Choices:

A. Neutrophilic infiltration with central necrosis B. Reactive gliosis with vascular proliferation C. Glial scarring with fibrous tissue hypertrophy D. Eosinophilic neuronal cytoplasm with pyknotic nuclei

Correct Answer: D. Eosinophilic neuronal cytoplasm with pyknotic nuclei

Question 81

Clinical Scenario:

A newlywed couple comes to your office for genetic counseling. Both potential parents are known to be carriers of the same Cystic Fibrosis (CF) mutation. What is the probability that at least one of their next three children will have CF if they are all single births?

Answer Choices:

A. 0 B. 1/64 C. 27/64 D. 37/64

Correct Answer: D. 37/64

Question 82

Clinical Scenario:

A 35-year-old soldier is rescued from a helicopter crash in the Arctic Circle and brought back to a treatment facility at a nearby military base. On arrival, the soldier's wet clothes are removed. He appears pale and is not shivering. The patient is unresponsive to verbal or painful stimuli. His temperature is 27.4°C (81.3°F), the pulse is 30/min and irregular, the respiratory rate is 7/min, and the blood pressure is 83/52 mm Hg. Examination shows fixed, dilated pupils, and diffuse rigidity. The fingers and toes are white in color and hard to touch. An ECG shows atrial fibrillation. In addition to emergent intubation, which of the following is the most appropriate next step in patient management?

Answer Choices:

A. Application of heating pads to the extremities B. Emergent electrical cardioversion C. Intravenous administration of tissue plasminogen activator D. Intravenous administration of warmed normal saline

Correct Answer: D. Intravenous administration of warmed normal saline

Question 83

Clinical Scenario:

A 17-year-old woman with no significant past medical history presents to the outpatient OB/GYN clinic with her parents for concerns of primary amenorrhea. She denies any symptoms and appears relatively unconcerned about her presentation. The review of systems is negative. Physical examination demonstrates an age-appropriate degree of development of secondary sexual characteristics, and no significant abnormalities on heart, lung, or abdominal examination. Her vital signs are all within normal limits. Her parents are worried and request that the appropriate laboratory tests are ordered. Which of the following tests is the best next step in the evaluation of this patient's primary amenorrhea?

Answer Choices:

A. Pelvic ultrasound B. Left hand radiograph C. Serum beta hCG D. Serum FSH

Correct Answer: C. Serum beta hCG

Question 84

Clinical Scenario:

A 4-year old boy is brought to the emergency department with fever, painful swallowing, headache, and neck spasm that began shortly after waking up. He has had a sore throat over the last week that acutely worsened

this morning. He has no history of serious illness and takes no medications. He lives at home with his mother. His older brother has asthma. His immunizations are up-to-date. He appears acutely ill. His temperature is 38.4°C (101.2°F), pulse is 95/min, respirations are 33/min, and blood pressure is 93/60 mm Hg. Examination shows drooling. The neck is stiff and extension is limited. Respirations appear labored with accessory muscle use. Inspiratory stridor is heard on auscultation of the chest. Cardiac examination shows no abnormalities. Oropharyngeal examination shows a bulge in the posterior pharyngeal wall. Intravenous access is obtained and laboratory studies are ordered. Which of the following is the most appropriate next step in the management of this patient?

Answer Choices:

A. Endotracheal intubation B. IV antibiotics C. Blood cultures D. IV corticosteroids

Correct Answer: A. Endotracheal intubation

Question 85

Clinical Scenario:

An investigator is studying the genetic profile of an isolated pathogen that proliferates within macrophages. The pathogen contains sulfatide on the surface of its cell wall to prevent fusion of the phagosome and lysosome. She finds that some of the organisms under investigation have mutations in a gene that encodes the enzyme required for synthesis of RNA from a DNA template. The mutations are most likely to reduce the therapeutic effect of which of the following drugs?

Answer Choices:

A. Streptomycin B. Rifampin C. Pyrazinamide D. Levofloxacin

Correct Answer: B. Rifampin

Question 86

Clinical Scenario:

A 33-year-old man comes to the physician for evaluation of progressive hair loss from his scalp. He first noticed receding of the hairline over the bitemporal regions of his scalp 5 years ago. Since then, his hair has gradually become thinner over the crown of his head. He is otherwise healthy and takes no medications. Examination shows diffuse, nonscarring hair loss over the scalp with a bitemporal pattern of recession. Administration of which of the following drugs is most appropriate to treat this patient's hair loss?

Answer Choices:

A. Clomipramine B. Triamcinolone C. Levothyroxine D. Finasteride

Correct Answer: D. Finasteride

Question 87

Clinical Scenario:

A 24-year-old man comes to the physician for a routine health maintenance examination. He feels well. He has type 1 diabetes mellitus. His only medication is insulin. He immigrated from Nepal 2 weeks ago. He lives in a shelter. He has smoked one pack of cigarettes daily for the past 5 years. He has not received any routine childhood vaccinations. The patient appears healthy and well nourished. He is 172 cm (5 ft 8 in) tall and weighs 68 kg (150 lb); BMI is 23 kg/m². His temperature is 36.8°C (98.2°F), pulse is 72/min, and blood pressure is 123/82 mm Hg. Examination shows a healed scar over his right femur. The remainder of the examination shows no abnormalities. A purified protein derivative (PPD) skin test is performed. Three days later, an induration of 13 mm is noted. Which of the following is the most appropriate initial step in the management of this patient?

Answer Choices:

A. Perform interferon-γ release assay B. Obtain a chest x-ray C. Administer isoniazid for 9 months D. Collect sputum sample for culture

Correct Answer: B. Obtain a chest x-ray

Question 88

Clinical Scenario:

A 3-year-old boy is brought to the physician because of recurrent nosebleeds and fatigue for the past 2 months. He also frequently complains his head hurts. The patient has met all motoric milestones for his age but does not like to run because his legs start to hurt if he does. He is at the 40th percentile for both height and weight. His temperature is 37.0°C (98.6°F), pulse is 125/min, respirations are 32/min, and blood pressure in the right arm is 130/85 mm Hg. A grade 2/6 systolic murmur is heard in the left paravertebral region. Further evaluation of this patient is most likely to show which of the following findings?

Answer Choices:

A. Inferior rib notching B. Pulmonary valve stenosis C. Left-axis deviation on ECG D. Delayed pulse in lower extremities

Correct Answer: D. Delayed pulse in lower extremities

Question 89

Clinical Scenario:

A 21-year-old man presents to the emergency room requesting surgery to remove "microchips," which he believes were implanted in his brain by "Russian spies" 6 months ago to control his thoughts. He also reports hearing the "spies" talk to each other through embedded "microspeakers." You notice that his hair appears

unwashed and some of his clothes are on backward. Urine toxicology is negative for illicit drugs. Which of the following additional findings are you most likely to see in this patient during the course of his illness?

Answer Choices:

A. Amnesia, multiple personality states, and de-realization B. Anhedonia, guilty rumination, and insomnia C. Asociality, flat affect, and alogia D. Grandiose delusions, racing thoughts, and pressured speech

Correct Answer: C. Asociality, flat affect, and alogia

Question 90

Clinical Scenario:

A 27-year-old woman is admitted to the emergency room with dyspnea which began after swimming and progressed gradually over the last 3 days. She denies cough, chest pain, or other respiratory symptoms. She reports that for the past 4 months, she has had several dyspneic episodes that occurred after the exercising and progressed at rest, but none of these were as long as the current one. Also, she notes that her tongue becomes 'wadded' when she speaks and she tires very quickly during the day. The patient's vital signs are as follows: blood pressure 125/60 mm Hg, heart rate 92/min, respiratory rate 34/min, and body temperature 36.2°C (97.2°F). Blood saturation on room air is initially 92% but falls to 90% as she speaks up. On physical examination, the patient is slightly lethargic. Her breathing is rapid and shallow. Lung auscultation, as well as cardiac, and abdominal examinations show no remarkable findings. Neurological examination reveals slight bilateral ptosis increased by repetitive blinking, and easy fatigability of muscles on repeated movement worse on the face and distal muscles of the upper and lower extremities. Which arterial blood gas parameters would you expect to see in this patient?

Answer Choices:

A. PaCO₂ = 34 mm Hg, PaO₂ = 61 mm Hg B. PaCO₂ = 31 mm Hg, PaO₂ = 67 mm Hg C. PaCO₂ = 51 mm Hg, PaO₂ = 58 mm Hg D. PaCO₂ = 37 mm Hg, PaO₂ = 46 mm Hg

Correct Answer: C. PaCO₂ = 51 mm Hg, PaO₂ = 58 mm Hg

Question 91

Clinical Scenario:

A 76-year-old woman with a history of hypertension and type 2 diabetes mellitus is brought to the emergency department 60 minutes after the acute onset of left-sided abdominal pain and nausea with vomiting. Three weeks ago, she underwent emergency surgical revascularization for acute left lower extremity ischemia. Physical examination shows left upper quadrant tenderness without rebound or guarding. Serum studies show an elevated lactate dehydrogenase level. Laboratory studies, including a complete blood count, basic metabolic panel, and hepatic panel, are otherwise unremarkable. A transverse section of a CT scan of the abdomen is shown. Further evaluation is most likely to show which of the following?

Answer Choices:

A. Absent P waves on electrocardiogram B. Non-compressible femoral vein on ultrasonography C. Infrarenal aortic aneurysm on abdominal CT scan D. Schistocytes on peripheral blood smear

Correct Answer: A. Absent P waves on electrocardiogram

Question 92

Clinical Scenario:

A 7-year-old boy is brought into the emergency department after he was found at home by his mother possibly drinking bleach from under the sink. The child consumed an unknown amount and appears generally well. The child has an unremarkable past medical history and is not currently taking any medications. Physical exam reveals a normal cardiopulmonary and abdominal exam. Neurological exam is within normal limits and the patient is cooperative and scared. The parents state that the ingestion happened less than an hour ago. Which of the following is the best next step in management?

Answer Choices:

A. Close observation and outpatient endoscopy in 2 to 3 weeks B. Nasogastric tube C. Titrate the alkali ingestion with a weak acid D. Urgent endoscopy

Correct Answer: A. Close observation and outpatient endoscopy in 2 to 3 weeks

Question 93

Clinical Scenario:

A 43-year-old man presents with a severe, throbbing, left-sided headache for the last 2 hours. He says that the pain has been progressively worsening and is aggravated by movement. The patient says he has had similar episodes in the past and would take acetaminophen and 'sleep it off'. He also complains that the light in the room is intolerably bright, and he is starting to feel nauseous. No significant past medical history and no current medications. Vital signs include: pulse 110/min, respiratory rate 15/min, and blood pressure 136/86 mm Hg. Physical examination reveals mild conjunctival injection in the left eye. Intraocular pressure (IOP) is normal. The rest of the examination is unremarkable. The patient is given a medication which relieves his symptoms. During discharge, he wants more of this medication to prevent episodes in future but he is told that the medication is only effective in terminating acute attacks but not for prevention. Which of the following receptors does the drug given to this patient bind to?

Answer Choices:

A. 5-hydroxytryptamine type 1 (5-HT₁) receptors B. Angiotensin II receptors C. 5-hydroxytryptamine type 2 (5-HT₂) receptors D. Muscarinic receptors

Correct Answer: A. 5-hydroxytryptamine type 1 (5-HT₁) receptors

Question 94

Clinical Scenario:

A 50-year-old man visits his primary care practitioner for a general health check-up. He was recently hired as a fitness instructor at a local fitness center. His father died of advanced colorectal cancer, however, his personal medical history is significant for the use of performance-enhancing drugs during his 20's when he competed in bodybuilding and powerlifting competitions. As part of the paperwork associated with his new position, he received an order for a hemoglobin and hematocrit, occult blood in stool, and serum iron and ferritin level, shown below: Hemoglobin 11.8 g/dL Hematocrit 35% Iron 40 µg/dL Ferritin 8 ng/mL His fecal occult blood test was positive. Which of the following is the most recommended follow-up action?

Answer Choices:

A. Endoscopy only B. Colonoscopy only C. Endoscopy and colonoscopy D. Transfusion

Correct Answer: C. Endoscopy and colonoscopy

Question 95

Clinical Scenario:

A 58-year-old man presents with a high-grade fever, throbbing left-sided headache, vision loss, and left orbital pain. He says that his symptoms started acutely 2 days ago with painful left-sided mid-facial swelling and a rash, which progressively worsened. Today, he woke up with complete vision loss in his left eye. His past medical history is significant for type 2 diabetes mellitus, diagnosed 5 years ago. He was started on an oral hypoglycemic agent which he discontinued after a year. His temperature is 38.9°C (102.0°F), blood pressure is 120/80 mm Hg, pulse is 120/min, and respiratory rate is 20/min. On examination, there is purulent discharge from the left eye and swelling of the left half of his face including the orbit. Oral examination reveals extensive necrosis of the palate with a black necrotic eschar and purulent discharge. Ophthalmic examination is significant for left-sided ptosis, proptosis, and an absence of the pupillary light reflex. Laboratory findings are significant for a blood glucose level of 388 mg/dL and a white blood cell count of 19,000 cells/mm³. Urinary ketone bodies are positive. Fungal elements are found on a KOH mount of the discharge. Which of the following statements best describes the organism responsible for this patient's condition?

Answer Choices:

A. It produces conidiospores B. It appears as a narrow-based budding yeast with a thick capsule C. Histopathological examination shows non-septate branching hyphae D. Histopathological examination shows acute angle branching hyphae

Correct Answer: C. Histopathological examination shows non-septate branching hyphae

Question 96

Clinical Scenario:

A 28-year-old man comes to the physician for the evaluation of a progressively worsening tremor in his hands and multiple falls over the past 3 months. The tremor occurs both at rest and with movement. He also reports decreased concentration and a loss of interest in his normal activities over this time period. He has no history of serious medical illness and takes no medications. He drinks two alcoholic beverages daily and does not use illicit drugs. Vital signs are within normal limits. Physical exam shows mild jaundice, a flapping tremor, and a broad-based gait. Serum studies show: Aspartate aminotransferase 554 U/L Hepatitis B surface antibody positive Hepatitis B surface antigen negative Ceruloplasmin 5.5 mg/dL (normal: 19.0-31.0 mg/dL) Which of the following is the most appropriate pharmacotherapy for this patient?"

Answer Choices:

A. Prednisolone B. Levodopa C. Deferoxamine D. Penicillamine

Correct Answer: D. Penicillamine

Question 97

Clinical Scenario:

A 65-year-old African-American man comes to the physician for a follow-up examination after presenting with elevated blood pressure readings during his last visit. He has no history of major medical illness and takes no medications. He is 180 cm (5 ft 9 in) tall and weighs 68 kg (150 lb); BMI is 22 kg/m². His pulse is 80/min and blood pressure is 155/90 mm Hg. Laboratory studies show no abnormalities. Which of the following is the most appropriate initial pharmacotherapy for this patient?

Answer Choices:

A. Metoprolol B. Chlorthalidone C. Aliskiren D. Captopril

Correct Answer: B. Chlorthalidone

Question 98

Clinical Scenario:

A 42-year-old man is brought to the emergency department by his wife because of a 1-day history of progressive confusion. He recently lost his job. He has a history of chronic alcoholism and has been drinking 14 beers daily for the past week. Before this time, he drank 6 beers daily. He appears lethargic. His vital signs are within normal limits. Serum studies show a sodium level of 111 mEq/L and a potassium level of 3.7 mEq/L. Urgent treatment for this patient's current condition increases his risk for which of the following adverse events?

Answer Choices:

A. Wernicke encephalopathy B. Cerebral edema C. Osmotic myelinolysis D. Hyperglycemia

Correct Answer: C. Osmotic myelinolysis

Question 99

Clinical Scenario:

A 59-year-old patient with COPD is admitted with difficulty breathing and increased sputum production. Approx. a week ago, he developed an upper respiratory tract infection. On admission, his blood pressure is 130/80 mm Hg, the heart rate 92/min, the respiratory rate 24/min, the temperature 37.6°C (99.7°F), and SaO₂ on room air 87%. Chest radiograph shows consolidation in the lower lobe of the right lung. Arterial blood gases (ABG) are taken and antibiotics are started. A nasal cannula provides 2L of oxygen to the patient. When the ABG results arrive, the patient's SaO₂ is 93%. The results are as follows: pH 7.32 PaO₂ 63 mm Hg PaCO₂ 57 mm Hg HCO₃⁻ 24 mEq/L What is the most appropriate next step in the management of this patient?

Answer Choices:

A. Increase oxygen to SaO₂ > 95% B. Start non-invasive positive pressure ventilation C. Intubate and start invasive ventilation D. Administer oral corticosteroids

Correct Answer: B. Start non-invasive positive pressure ventilation

Question 100

Clinical Scenario:

A 28-year-old G1P0 woman who is 30 weeks pregnant presents to the women's health center for a prenatal checkup. She is concerned that her baby is not moving as much as usual over the past five days. She thinks she only felt the baby move eight times over an hour long period. Her prenatal history was notable for morning sickness requiring pyridoxine. Her second trimester ultrasound revealed no abnormal placental attachment. She takes a multivitamin daily. Her temperature is 98.6°F (37°C), blood pressure is 120/70 mmHg, pulse is 80/min, and respirations are 16/min. The patient's physical exam is unremarkable. Her fundal height is 28 cm, compared to 26 cm two weeks ago. The fetal pulse is 140/min. The patient undergoes external fetal monitoring. With vibroacoustic stimulation, the patient feels eight movements over two hours. What is the best next step in management?

Answer Choices:

A. Induction of labor B. Oxytocin challenge C. Biophysical profile D. Inpatient monitoring

Correct Answer: C. Biophysical profile

Question 101

Clinical Scenario:

A 45-year-old man presents with a persistent cough for the past month. He says it started off with a runny nose and fever, from which he recovered in a week, but he says that the cough persists after the resolution of the fever. The patient denies any expectoration, chest pain, weight loss, or breathlessness. He reports no history of recent travel or sick contacts. Past medical history is significant for chronic constipation. He reports a 15-pack-

year smoking history but denies any alcohol or current recreational drug use. He says he did use intravenous drugs in his late twenties but quit after going through a drug rehabilitation program. Physical examination is unremarkable. Laboratory findings and a chest radiograph are normal. Which of the following would be the best choice as a cough suppressant in this patient?

Answer Choices:

A. Codeine B. Dextromethorphan C. Pseudoephedrine D. Oxymetazoline

Correct Answer: B. Dextromethorphan

Question 102

Clinical Scenario:

A 3-year-old boy is brought to the physician for presurgical evaluation before undergoing splenectomy. One year ago, he was diagnosed with hereditary spherocytosis and has received 6 blood transfusions for severe anemia since then. His only medication is a folate supplement. Immunizations are up-to-date. His temperature is 36.7°C (98°F), pulse is 115/min, respirations are 24/min, and blood pressure is 110/60 mm Hg. Examination shows conjunctival pallor and jaundice. The spleen tip is palpated 5 cm below the left costal margin. Which of the following is the most appropriate recommendation to prevent future morbidity and mortality in this patient?

Answer Choices:

A. Vaccination against hepatitis B virus B. Daily penicillin prophylaxis C. Daily warfarin prophylaxis D. Administration of hydroxyurea

Correct Answer: B. Daily penicillin prophylaxis

Question 103

Clinical Scenario:

A 16-year-old college student presents to the emergency department with a 3-day history of fever, muscle rigidity, and confusion. He was started on a new medication for schizophrenia 2 months ago. There is no history of sore throat, burning micturition, or loose motions. At the hospital, his temperature is 38.6°C (101.5°F); the blood pressure is 108/62 mm Hg; the pulse is 120/min, and the respiratory rate is 16/min. His urine is cola-colored. On physical examination, he is sweating profusely. Treatment is started with antipyretics and intravenous hydration. Which of the following is most likely responsible for this patient's condition?

Answer Choices:

A. Chlorpromazine B. Diazepam C. Levodopa D. Phenytoin

Correct Answer: A. Chlorpromazine

Question 104

Clinical Scenario:

A 28-year-old man is brought to the emergency department by ambulance after being hit in the head with a baseball bat. Physical examination shows swelling and bruising around the left temple and eye. A CT scan of the head shows a transverse fracture through the sphenoid bone and blood in the sphenoid sinus. Neurological examination is most likely to show which of the following findings?

Answer Choices:

A. Inward deviation of the left eye B. Left facial paralysis C. Decreased hearing in the left ear D. Deviation of uvula to the right

Correct Answer: A. Inward deviation of the left eye

Question 105

Clinical Scenario:

An 8-year-old boy is brought to the physician because of worsening confusion and lethargy for the last hour. He has had high-grade fever, productive cough, fatigue, and malaise for 2 days. He was diagnosed with sickle cell anemia at the age of 2 years but has not seen a physician in over a year. His temperature is 38.9°C (102°F), pulse is 133/min, respirations are 33/min, and blood pressure is 86/48 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 92%. The patient does not respond to verbal commands. Examination shows conjunctival pallor and scleral icterus. Inspiratory crackles are heard at the left lung base. Laboratory studies show: Hemoglobin 8.1 g/dL Leukocyte count 17,000/mm³ Platelet count 200,000/mm³ Which of the following is most likely to have prevented this patient's condition?"

Answer Choices:

A. Chronic transfusion therapy B. Polysaccharide vaccination C. Folic acid D. Low molecular weight heparin

Correct Answer: B. Polysaccharide vaccination

Question 106

Clinical Scenario:

A 64-year-old man is brought to the emergency department because of dull lower abdominal pain for 3 hours. He has not urinated for 24 hours and has not passed stool for over 3 days. He was diagnosed with herpes zoster 4 weeks ago and continues to have pain even after his rash resolved. He has hypertension, benign prostatic hyperplasia, and coronary artery disease. Physical examination shows a tender, palpable suprapubic mass. Bowel sounds are hypoactive. Abdominal ultrasound shows a large anechoic mass in the pelvis. Which of the following drugs most likely accounts for this patient's current symptoms?

Answer Choices:

A. Simvastatin B. Amlodipine C. Valproate D. Desipramine

Correct Answer: D. Desipramine

Question 107

Clinical Scenario:

A 23-year-old man comes to the physician because of a tremor in his right hand for the past 3 months. The tremor has increased in intensity and he is unable to perform his daily activities. When he wakes up in the morning, his pillow is soaked in saliva. During this period, he has been unable to concentrate in his college classes. He has had several falls over the past month. He has no past history of serious illness. He appears healthy. His vital signs are within normal limits. Examination shows a broad-based gait. There is a low frequency tremor that affects the patient's right hand to a greater extent than his left. When the patient holds his arms fully abducted with his elbows flexed, he has a bilateral low frequency arm tremor that increases in amplitude the longer he holds his arms up. Muscle strength is normal in all extremities. Sensation is intact. Deep tendon reflexes are 4+ bilaterally. Dysmetria is present. A photograph of the patient's eye is shown. Mental status examination shows a restricted affect. The rate and rhythm of his speech is normal. Which of the following is the most appropriate pharmacotherapy?

Answer Choices:

A. Penicillamine B. Deferoxamine C. Prednisone D. Levodopa "

Correct Answer: A. Penicillamine

Question 108

Clinical Scenario:

A 27-year-old man comes to the physician for a follow-up examination. Paroxetine therapy was initiated 6 weeks ago for a major depressive episode. He now feels much better and says he is delighted with his newfound energy. He gets around 8 hours of sleep nightly. His appetite has increased. Last year, he had two episodes of depressed mood, insomnia, and low energy during which he had interrupted his job training and stopped going to the gym. Now, he has been able to resume his job at a local bank. He also goes to the gym three times a week to work out and enjoys reading books again. His temperature is 36.5°C (97.7°F), pulse is 70/min, and blood pressure is 128/66 mm Hg. Physical and neurologic examinations show no abnormalities. On mental status examination, he describes his mood as ""good." Which of the following is the most appropriate next step in management?"

Answer Choices:

A. Continue paroxetine therapy for 2 years B. Discontinue paroxetine C. Continue paroxetine therapy for 6 months D. Switch from paroxetine to lithium therapy "

Correct Answer: A. Continue paroxetine therapy for 2 years

Question 109

Clinical Scenario:

A 12-year-old boy is brought in by his mother for a routine checkup. The patient's mother says he is frequently fatigued and looks pale. She also claims that he has recently become "much quieter" than normal and is no longer interested in playing baseball with his friends. The patient's mother believes it may just be "growing pains." The patient has no significant medical history. He is the 90th percentile for height and weight and has been meeting all developmental milestones. The patient is afebrile, and his vital signs are within normal limits. Physical examination reveals several small bruises on the patient's right arm and on both thighs. Laboratory findings are significant for the following: Sodium 140 mEq/L Potassium 4.2 mEq/L Chloride 101 mEq/L Bicarbonate 27 mEq/L BUN 16 mg/dL Creatinine 1.2 mg/dL Glucose (fasting) 111 mg/dL WBC 3,400/mm³ RBC 4.20 x 10⁶/mm³ Hematocrit 22% Hemoglobin 7.1 g/dL Platelet count 109,000/mm³ A peripheral blood smear reveals myeloblasts. Which of the following is the next best step in the management of this patient?

Answer Choices:

A. Referral to social services B. Administration of oral ferrous sulfate C. Bone marrow biopsy D. Chest radiograph

Correct Answer: C. Bone marrow biopsy

Question 110

Clinical Scenario:

An 18-year-old woman presents to the emergency department with severe right lower quadrant discomfort and stomach pain for the past day. She has no significant past medical history. She states that she is sexually active and uses oral contraceptive pills for birth control. Her vital signs include: blood pressure 127/81 mm Hg, pulse 101/min, respiratory rate 19/min, and temperature 39.0°C (102.2°F). Abdominal examination is significant for focal tenderness and guarding in the right lower quadrant. Blood is drawn for lab tests which reveal the following: Hb% 13 gm/dL Total count (WBC) 15,400 /mm³ Differential count Neutrophils: Segmented 70% Band Form 5% Lymphocytes 20% Monocytes 5% What is the next best step in the management of this patient?

Answer Choices:

A. Pelvic exam B. Ultrasound of the pelvis C. Ultrasound of the appendix D. Upper gastrointestinal series

Correct Answer: C. Ultrasound of the appendix

Question 111

Clinical Scenario:

A 36-year-old woman comes to the physician to discuss contraceptive options. She is currently sexually active with one male partner, and they have not been using any contraception. She has no significant past medical

history and takes no medications. She has smoked one pack of cigarettes daily for 15 years. She is allergic to latex and copper. A urine pregnancy test is negative. Which of the following contraceptive methods is contraindicated in this patient?

Answer Choices:

A. Diaphragm with spermicide B. Progestin-only pill C. Intrauterine device D. Combined oral contraceptive pill

Correct Answer: D. Combined oral contraceptive pill

Question 112

Clinical Scenario:

A 65-year-old man comes to the physician for evaluation of a neck mass and weight loss. He first noticed the growing mass 2 months ago. The mass is not painful. He also has decreased appetite and intermittent abdominal pain. He has lost 10 kg (22 lb) of weight over the past 3 months. Sometimes, he wakes up in the morning drenched in sweat. He takes daily over-the-counter multivitamins. He appears pale. His pulse is 65/min, blood pressure is 110/70 mm Hg, and temperature is 38.1°C (100.6°F). Physical exam shows a painless, golf ball-sized mass in the anterior triangle of the neck. A biopsy shows large cells with a bilobed nucleus that are CD15- and CD30-positive. Laboratory analysis of serum shows a calcium level of 14.5 mg/dL and a parathyroid hormone level of 40 pg/mL. Which of the following is the most likely explanation of this patient's laboratory findings?

Answer Choices:

A. Osteoblastic metastasis B. Ectopic vitamin D production C. Multivitamin overdose D. Osteolytic metastasis

Correct Answer: B. Ectopic vitamin D production

Question 113

Clinical Scenario:

A 27-year-old primigravid woman at 32 weeks' gestation comes to the physician for a prenatal visit. She has had swollen legs, mild shortness of breath, and generalized fatigue for the past 2 weeks. Medications include iron supplements and a multivitamin. Her temperature is 37.2°C (99°F), pulse is 93/min, respirations are 20/min, and blood pressure is 108/60 mm Hg. There is 2+ pitting edema of the lower extremities, but no erythema or tenderness. The lungs are clear to auscultation. Cardiac examination shows an S3 gallop. Pelvic examination shows a uterus consistent in size with a 32-week gestation. Which of the following is the most appropriate next step in management for this patient's symptoms?

Answer Choices:

A. Urinalysis B. Lower extremity doppler C. Ventilation-perfusion scan D. Reassurance and monitoring "

Correct Answer: D. Reassurance and monitoring "

Question 114

Clinical Scenario:

A 45-year-old man presents with a long history of ulcers on the bottom of his feet. He recalls having a similar looking ulcer on the side of his penis when he was 19 years old for which he never sought treatment. The patient denies any fever, chills, or constitutional symptoms. He reports multiple sexual partners and a very promiscuous sexual history. He has also traveled extensively as a writer since he was 19. The patient is afebrile, and his vital signs are within normal limits. A rapid plasma reagin (RPR) test is positive, and the result of a Treponema pallidum particle agglutination (TP-PA) is pending. Which of the following findings would most likely be present in this patient?

Answer Choices:

A. Wide-based gait with a low step B. Positive Romberg's sign C. Memory loss D. Agrophesthesia

Correct Answer: B. Positive Romberg's sign

Question 115

Clinical Scenario:

A 45-year-old woman, gravida 3, para 2, at 18 weeks' gestation comes to the physician for a prenatal visit. Ultrasonography at a previous visit when she was at 12 weeks' gestation showed a hypoplastic nasal bone. Pelvic examination shows a uterus consistent in size with an 18-week gestation. Maternal serum studies show low α -fetoprotein and free estriol concentrations, and increased inhibin A and β -hCG concentrations. Physical examination of the infant after delivery is most likely to show which of the following findings?

Answer Choices:

A. Ambiguous external genitalia B. Extremity lymphedema C. Meningomyelocele D. Single transverse palmar crease

Correct Answer: D. Single transverse palmar crease

Question 116

Clinical Scenario:

A 72-year-old man with a 4-year history of Parkinson disease comes to the physician for evaluation of his medication. Since his last visit one year ago, he has had increased tremor and bradykinesia up to an hour before his next scheduled dose and sometimes feels like he does not respond to some doses at all. One week ago, he was entirely unable to move for about a minute when he wanted to exit an elevator. The physician prescribes a drug that increases the bioavailability of levodopa by preferentially preventing its peripheral methylation. This patient was most likely prescribed which of the following drugs by the physician?

Answer Choices:

A. Entacapone B. Ropinirole C. Amantadine D. Rasagiline

Correct Answer: A. Entacapone

Question 117

Clinical Scenario:

A previously healthy 6-week-old infant is brought to the emergency department because of fever, fatigue, and dry cough for one day. She has been feeding poorly and had difficulty latching on to breastfeed since yesterday. She has had nasal congestion. The mother reports that her daughter has not been going through as many diapers as usual. She was born by uncomplicated vaginal delivery at 42 weeks' gestation. Her mother is a cystic fibrosis carrier. The patient has been treated with acetaminophen for the last 24 hours, and vitamin D drops since birth. She appears irritable, pale, and lethargic. She is at the 25th percentile for both length and weight; she had the same percentiles at birth. Her temperature is 38.2°C (100.7°F) and respirations are 64/min. Pulse oximetry on room air shows an oxygen saturation of 92%. Examination shows an ill-appearing infant with a cough and nasal flaring. Mucous membranes are dry. Chest examination shows intercostal and supraclavicular retractions. Expiratory wheezes are heard on auscultation. Which of the following is the most likely causal organism?

Answer Choices:

A. *Listeria monocytogenes* B. Respiratory syncytial virus C. Coronavirus D. *Streptococcus pneumoniae*

Correct Answer: B. Respiratory syncytial virus

Question 118

Clinical Scenario:

A 1-day-old infant presents to the office because the mother noticed "blood" in the diaper of her child. She has brought the diaper with her which shows a small reddish marking. The pregnancy was without complications, as was the delivery. The patient presents with no abnormal findings on physical examination. The laboratory analysis shows uric acid levels in the blood to be 5 mg/dL. Which of the following should be the next step in management?

Answer Choices:

A. Allopurinol B. Febuxostat C. No therapy is required D. Sodium bicarbonate

Correct Answer: C. No therapy is required

Question 119

Clinical Scenario:

A 6-year-old boy is brought to the physician by his parents because of right lower extremity weakness, worsening headaches, abdominal pain, dark urine, and a 5-kg (11-lb) weight loss for the past 2 months. His teachers report that he has not been paying attention in class and his grades have been worsening. He has a

history of infantile seizures. Physical examination shows a palpable abdominal mass and left costovertebral angle tenderness. Neurological exam shows decreased strength of the right lower limb. He has several acne-like angiofibromas around the nose and cheeks. Further evaluation is most likely to show which of the following?

Answer Choices:

A. Port wine stain B. Lisch nodules C. Subependymal giant cell astrocytoma D. Vestibular schwannoma

Correct Answer: C. Subependymal giant cell astrocytoma

Question 120

Clinical Scenario:

A 12-year-old girl is brought to the physician by her mother because of high fever and left ankle and knee joint swelling. She had a sore throat 3 weeks ago. There is no family history of serious illness. Her immunizations are up-to-date. She had an episode of breathlessness and generalized rash when she received dicloxacillin for a skin infection 2 years ago. She appears ill. Her temperature is 38.8°C (102.3°F), pulse is 87/min, and blood pressure is 98/62 mm Hg. Examination shows left ankle and knee joint swelling and tenderness; range of motion is limited. Breath sounds over both lungs are normal. A grade 3/6 holosystolic murmur is heard best at the apex. Abdominal examination is normal. Which of the following is the most appropriate pharmacotherapy?

Answer Choices:

A. Clarithromycin B. High-dose glucocorticoids C. Amoxicillin D. Methotrexate

Correct Answer: A. Clarithromycin

Question 121

Clinical Scenario:

A 32-year-old man is brought to the emergency department by the police for examination. The police have reason to believe he may have swallowed a large number of cocaine-containing capsules during an attempt to smuggle the drug across the border. They request an examination of the patient to determine if this is actually the case. The patient has no history of any serious illnesses and takes no medications. He does not smoke, drinks, or consume any drugs. He appears upset. His vital signs are within normal limits. Despite the pressure by the police, he refuses to undergo any further medical evaluation. Which of the following is the most appropriate next step in the evaluation of this patient?

Answer Choices:

A. Examine the patient without his consent B. Explain the risk of internal rupture to the patient C. Obtain an abdominal X-ray D. Request a court order from the police

Correct Answer: B. Explain the risk of internal rupture to the patient

Question 122

Clinical Scenario:

A 19-year-old woman comes to the physician because of a 1-year history of severe abdominal pain, bloating, and episodic diarrhea. She has also has a 10-kg (22-lb) weight loss over the past 10 months. Physical examination shows a mildly distended abdomen, diffuse abdominal tenderness, and multiple erythematous, tender nodules on the anterior aspect of both legs. There is a small draining lesion in the perianal region. Further evaluation of this patient's gastrointestinal tract is most likely to show which of the following findings?

Answer Choices:

A. Villous atrophy B. No structural abnormalities C. Melanosis coli D. Transmural inflammation

Correct Answer: D. Transmural inflammation

Question 123

Clinical Scenario:

A 30-year-old African American woman comes to the physician because of fatigue and muscle weakness for the past 5 weeks. During this period, she has had recurrent headaches and palpitations. She has hypertension and major depressive disorder. She works as a nurse at a local hospital. She has smoked about 6–8 cigarettes daily for the past 10 years and drinks 1–2 glasses of wine on weekends. Current medications include enalapril, metoprolol, and fluoxetine. She is 160 cm (5 ft 6 in) tall and weighs 60 kg (132 lb); BMI is 21.3 kg/m². Her temperature is 37°C (98.6°F), pulse is 75/min, and blood pressure is 155/85 mm Hg. The lungs are clear to auscultation. Cardiac examination shows no abnormalities. The abdomen is soft and nontender; bowel sounds are normal. Her skin is dry and there is no edema in the lower extremities. Laboratory studies show: Hemoglobin 13.3 g/dL Serum Na⁺ 146 mEq/L Cl⁻ 105 mEq/L K⁺ 3.0 mEq/L HCO₃⁻ 30 mEq/L Urea nitrogen 10 mg/dL Glucose 95 mg/dL Creatinine 0.8 mg/dL Urine Blood negative Glucose negative Protein negative RBC 0–1/hpf WBC none Which of the following is the most likely diagnosis in this patient?"

Answer Choices:

A. Laxative abuse B. Aldosteronoma C. Pheochromocytoma D. Cushing syndrome

Correct Answer: B. Aldosteronoma

Question 124

Clinical Scenario:

A 76-year-old woman presents to the physician for a follow-up examination. She had a hemoglobin level of 10.5 g/dL last month. She complains of mild dyspnea with exercise. She reports exercising daily for the past 30 years. She is relatively healthy without any significant past medical history. She occasionally takes ibuprofen for knee pain. She denies a prior history of alcohol or tobacco use. Her temperature is 37.1°C (98.8°F), the pulse is 65/min, the respiratory rate is 13/min, and the blood pressure is 115/65 mm Hg. The examination

shows no abnormalities. Laboratory studies show: Laboratory test Hemoglobin 10.5 g/dL Mean corpuscular volume 75 μm^3 Leukocyte count 6500/mm³ with a normal differential Platelet 400,000/mm³ Serum Iron 35 Total iron-binding capacity 450 $\mu\text{g/dL}$ Ferritin 8 Ca^{+} 9.0 mg/dL Albumin 3.9 g/dL Urea nitrogen 10 mg/dL Creatinine 0.9 mg/dL Serum protein electrophoresis and immunofixation show a monoclonal protein of 20 g/L (non-IgM). Marrow plasmacytosis is 5%. A skeletal survey shows no abnormalities. In addition to the workup of iron deficiency anemia, which of the following is the most appropriate next step in management?

Answer Choices:

A. Annual follow-up with laboratory tests B. Check beta-2 microglobulin C. Referral for radiation therapy D. No further steps are required at this time

Correct Answer: A. Annual follow-up with laboratory tests

Question 125

Clinical Scenario:

A 24-year-old woman is in the intensive care unit for the management of a severe acute asthma exacerbation. She is currently intubated and sedated, and she is receiving intravenous steroids, continuous nebulized beta-agonists, and anticholinergic therapy via breathing treatments. On hospital day 2, she has a new fever to 38.9°C (102.0°F). Chest X-ray shows a right lower lobe consolidation. Blood cultures are collected, and she is started empirically on intravenous cefepime and daptomycin. On hospital day 4, she continues to be febrile; chest X-ray shows interval worsening of the right lower lobe opacity. Which of the following is the most likely reason for treatment failure in this patient?

Answer Choices:

A. Abnormally rapid clearance of the medicines by the kidney B. Abnormally rapid metabolism of the medicines by the liver C. Inactivation of the medicine in the target tissue D. Low bioavailability of the medicines

Correct Answer: C. Inactivation of the medicine in the target tissue

Question 126

Clinical Scenario:

A 57-year-old woman presents to her primary care physician with complaints of nausea, vomiting, abdominal pain, and bloating that have increased in severity over the past several months. She reports that she occasionally vomits after eating. She states that the emesis contains undigested food particles. Additionally, the patient states that she often is satiated after only a few bites of food at meals. Her medical history is significant for hypertension and type II diabetes mellitus. Initial laboratory values are notable only for a hemoglobin A1c of 14%. Which of the following is the best initial treatment for this patient?

Answer Choices:

A. Dietary modification B. Erythromycin C. Myotomy D. Surgical resection

Correct Answer: A. Dietary modification

Question 127

Clinical Scenario:

A 62-year-old woman comes to the physician because of worsening mental status over the past month. Her husband reports that she was initially experiencing lapses in memory but has recently started having difficulties performing activities of daily living. She appears withdrawn and avoids eye contact. Examination shows diffuse involuntary muscle jerking that can be provoked by loud noises. A cerebrospinal fluid analysis shows elevated concentration of 14-3-3 protein. Four months later, the patient dies. Pathologic examination of the brain on autopsy is most likely to show which of the following findings?

Answer Choices:

A. Marked atrophy of caudate and putamen B. Focal inflammatory demyelination and gliosis C. Deposits of amyloid beta peptides D. Spongiform vacuolation of the cortex

Correct Answer: D. Spongiform vacuolation of the cortex

Question 128

Clinical Scenario:

A 3-week-old male infant is brought to the physician for follow-up. He was delivered at 30 weeks' gestation via Cesarean section and was cyanotic at birth, requiring resuscitation and a neonatal intensive care unit hospitalization. His mother received no prenatal care; she has diabetes mellitus type II and hypertension. She was not tested for sexually transmitted infections during the pregnancy. The infant appears well. Ophthalmologic examination shows tortuous retinal vessels. There are well-demarcated areas of non-vascularized retina in the periphery. This patient's retinal findings are most likely a result of which of the following?

Answer Choices:

A. Oxygen toxicity B. Glucocorticoid deficiency C. Hyperglycemia D. Syphilis infection

Correct Answer: A. Oxygen toxicity

Question 129

Clinical Scenario:

A 72-year-old man is brought to the emergency department with increasing fever and abdominal pain over the past week. The pain is constant and limited to the lower right part of his abdomen. He has nausea but no vomiting or diarrhea. His past medical history is unremarkable for any serious illnesses. He takes acetaminophen for knee arthritis. He is fully alert and oriented. His temperature is 39.5°C (103.1°F), pulse is

89/min, respirations are 15/min, and blood pressure is 135/70 mm Hg. Abdominal examination shows a tender mass in the right lower quadrant. CT shows obstruction of the appendiceal neck with a fecalith and the appendiceal tip leading to an irregular walled-off fluid collection. Stranding of the surrounding fat planes is also noted. Intravenous hydration is initiated. Which of the following is the most appropriate next step in management?

Answer Choices:

- A. Antibiotics + CT-guided drainage
- B. Antibiotics + interval appendectomy
- C. Appendectomy within 12 hours
- D. Early surgical drainage + interval appendectomy

Correct Answer: A. Antibiotics + CT-guided drainage

Question 130

Clinical Scenario:

A 17-year-old boy is brought to the pediatrician by his mother for an initial visit. He recently immigrated from Cambodia. Through an interpreter, the patient reports 6 months of mild exertional dyspnea. He denies chest pain or palpitations. His medical history is unremarkable and he has never had any surgeries. His family history is significant for hypertension and diabetes. His father died of tuberculosis. The patient's vaccination history is unknown. His temperature is 98°F (36.7°C), blood pressure is 113/71 mmHg, and pulse is 82/min. His BMI is 24 kg/m². Physical examination shows a well-nourished, cooperative boy without any grossly dysmorphic features. Cardiac auscultation reveals a grade II systolic ejection murmur along the left upper sternal border and a mid-diastolic rumble along the left sternal border. S1 is normal and the splitting of S2 does not change with inspiration. Which of the following is the most likely diagnosis?

Answer Choices:

- A. Atrial septal defect
- B. Bicuspid aortic valve
- C. Hypertrophic cardiomyopathy
- D. Ventricular septal defect

Correct Answer: A. Atrial septal defect

Question 131

Clinical Scenario:

A 37-year-old woman is being evaluated for difficulty with swallowing for the past few months. She explains that she experiences difficulty swallowing solid foods only. Her medical history is relevant for hypothyroidism and migraines. Her current medications include daily levothyroxine and acetaminophen as needed for pain. The vital signs include blood pressure 110/90 mm Hg, pulse rate 55/min, and respiratory rate 12/min. On physical examination, her abdomen is non-tender. Her voice is hoarse, but there is no pharyngeal hyperemia on oral examination. On cardiac auscultation, an opening snap followed by an early to mid-diastolic rumble is heard over the apex. A barium swallow X-ray is performed and is unremarkable. Echocardiography shows an enlarged left atrium and abnormal blood flow through 1 of the atrioventricular valves. What is the most likely valve abnormality seen in this patient?

Answer Choices:

A. Mitral valve stenosis B. Aortic valve stenosis C. Aortic valve regurgitation D. Mitral valve prolapse

Correct Answer: A. Mitral valve stenosis

Question 132

Clinical Scenario:

An investigator is studying the interaction between a new drug B and an existing drug A. The results are recorded and plotted on the graph shown. Which of the following properties of drug B best explain the observed effect on the dose-response curve of drug A?

Answer Choices:

A. Non-competitive antagonist B. Competitive antagonist C. Inverse agonist D. Functional antagonist

Correct Answer: B. Competitive antagonist

Question 133

Clinical Scenario:

A 40-year-old woman who works as a secretary presents to your office complaining of new pain and numbness in both of her hands. For the past few weeks, the sensation has occurred after long days of typing, but it now occasionally wakes her up from sleep. You do not note any deformities of her wrists or hands, but you are able to reproduce pain and numbness in the first three and a half digits by tapping the wrist. What is the best initial treatment for this patient's complaint?

Answer Choices:

A. A trial of gabapentin B. Local steroid injections C. Carpal tunnel release surgery D. Splinting

Correct Answer: D. Splinting

Question 134

Clinical Scenario:

A 68-year-old man presents to the emergency department with palpitations. He also feels that his exercise tolerance has reduced over the previous week. His past history is positive for ischemic heart disease and he has been on multiple medications for a long time. On physical examination, his temperature is 36.9°C (98.4°F), pulse rate is 152/min and is regular, blood pressure is 114/80 mm Hg, and respiratory rate is 18/min. Auscultation of the precordial region confirms tachycardia, but there is no murmur or extra heart sounds. His ECG is obtained, which suggests a diagnosis of atrial flutter. Which of the following findings is most likely to be present on his electrocardiogram?

Answer Choices:

A. Atrial rate above 400 beats per minute B. Slurred upstroke of R wave C. Atrioventricular block D. No discernible P waves

Correct Answer: C. Atrioventricular block

Question 135**Clinical Scenario:**

A 29-year-old woman presents to the primary care office for a recent history of falls. She has fallen 5 times over the last year. These falls are not associated with any preceding symptoms; she specifically denies dizziness, lightheadedness, or visual changes. However, she has started noticing that both of her legs feel weak. She's also noticed that her carpet feels strange beneath her bare feet. Her mother and grandmother have a history of similar problems. On physical exam, she has notable leg and foot muscular atrophy and 4/5 strength throughout her bilateral lower extremities. Sensation to light touch and pinprick is decreased up to the mid-calf. Ankle jerk reflex is absent bilaterally. Which of the following is the next best diagnostic test for this patient?

Answer Choices:

A. Ankle-brachial index B. Electromyography (including nerve conduction studies) C. Lumbar puncture D. MRI brain

Correct Answer: B. Electromyography (including nerve conduction studies)

Question 136**Clinical Scenario:**

A 28-year-old man is admitted to the emergency department with a gunshot wound to the abdomen. He complains of weakness and diffuse abdominal pain. Morphine is administered and IV fluids are started by paramedics at the scene. On admission, the patient's blood pressure is 90/60 mm Hg, heart rate is 103/min, respiratory rate is 17/min, the temperature is 36.2°C (97.1°F), and oxygen saturation is 94% on room air. The patient is responsive but lethargic. The patient is diaphoretic and extremities are pale and cool. Lungs are clear to auscultation. Cardiac sounds are diminished. Abdominal examination shows a visible bullet entry wound in the left upper quadrant (LUQ) with no corresponding exit wound on the flanks or back. The abdomen is distended and diffusely tender with a rebound. Aspiration of the nasogastric tube reveals bloody contents. Rectal examination shows no blood. Stool guaiac is negative. Which of the following is the next best step in management?

Answer Choices:

A. Exploratory laparotomy B. Focused assessment with sonography for trauma (FAST) C. Abdominal X-ray D. Diagnostic peritoneal lavage

Correct Answer: A. Exploratory laparotomy

Question 137

Clinical Scenario:

A 59-year-old man is brought to the emergency department 30 minutes after having a seizure. His wife reports that the patient has been having recurrent headaches and has become increasingly irritable over the past 3 months. Physical examination shows a spastic paresis of the right lower extremity. The Babinski sign is present on the right side. An MRI of the brain is shown. Which of the following is the most likely diagnosis?

Answer Choices:

A. Pituitary adenoma B. Ependymoma C. Oligodendroglioma D. Meningioma

Correct Answer: D. Meningioma

Question 138

Clinical Scenario:

A 20-year-old male is brought to a psychiatrist by his parents for bizarre behavior. His parents report that over the past two semesters in school, his personality and behavior have changed noticeably. He refuses to leave his room because he believes people are spying on him. He hears voices that are persecutory and is convinced that people at school have chips implanted in their brains to spy on him. Screenings for depression and mania are negative. His past medical history is unremarkable. His family history is notable for a maternal uncle with bipolar disorder. He does not drink alcohol or smoke. His temperature is 98.8°F (37.1°C), blood pressure is 115/70 mmHg, pulse is 85/min, and respirations are 18/min. On examination, he appears to be responding to internal stimuli. Which of the following pathways is primarily responsible for these symptoms?

Answer Choices:

A. Mesolimbic pathway B. Nigrostriatal pathway C. Tuberoinfundibular pathway D. Papez circuit

Correct Answer: A. Mesolimbic pathway

Question 139

Clinical Scenario:

A 23-year-old woman presents to a medical office for a check-up. The patient has a 5-year history of epilepsy with focal-onset motor seizures and currently is seizure-free on 50 mg of lamotrigine 3 times a day. She does not have any concurrent illnesses and does not take other medications, except oral contraceptive pills. She is considering pregnancy and seeks advice on possible adjustments or additions to her therapy. Which of the following changes should be made?

Answer Choices:

A. Decrease the dose of lamotrigine to 50 mg 2 times a day B. Recommend 5 mg of folic acid daily with no changes to antiepileptic therapy C. Recommend 100 µg of vitamin K daily with no changes to antiepileptic therapy D. No changes or additions to the patient's regimen are indicated

Correct Answer: B. Recommend 5 mg of folic acid daily with no changes to antiepileptic therapy

Question 140

Clinical Scenario:

A 62-year-old man presents to the emergency department with confusion. The patient's wife states that her husband has become more somnolent over the past several days and now is very confused. The patient has no complaints himself, but is answering questions inappropriately. The patient has a past medical history of diabetes and hypertension. His temperature is 98.3°F (36.8°C), blood pressure is 127/85 mmHg, pulse is 138/min, respirations are 14/min, and oxygen saturation is 99% on room air. Physical exam is notable for a confused man with dry mucous membranes. Initial laboratory studies are ordered as seen below.

Serum: Na⁺: 135 mEq/L Cl⁻: 100 mEq/L K⁺: 3.0 mEq/L HCO₃⁻: 23 mEq/L BUN: 30 mg/dL Glucose: 1,299 mg/dL Creatinine: 1.5 mg/dL Ca²⁺: 10.2 mg/dL

Which of the following is the most appropriate initial treatment for this patient?

Answer Choices:

A. Insulin B. Insulin and potassium C. Insulin, normal saline, and potassium D. Normal saline and potassium

Correct Answer: D. Normal saline and potassium

Question 141

Clinical Scenario:

A 17-year-old boy is brought to the physician by his parents who are concerned about his bizarre behavior. Over the past three months, he has become withdrawn from his friends and less interested in his classes and extracurricular activities. On several occasions, he has torn apart rooms in their home looking for "bugs" and states that the President is spying on him because aliens have told the government that he is a threat. Although he has always been quite clean in the past, his father notes that the patient's room is now malodorous with clothes and dishes strewn about haphazardly. He also says that sometimes he can hear the devil speaking to him from inside his head. He has no medical problems, does not drink alcohol or use any drugs. Physical examination of the boy reveals no abnormalities. On mental status examination, the boy is oriented to person, place and time. He avoids eye contact and replies mostly with monosyllabic responses. He appears distracted, and confirms that he is hearing whispering voices in his head. What is the most appropriate diagnosis for this patient?

Answer Choices:

A. Brief psychotic disorder B. Schizoaffective disorder C. Schizophreniform disorder D. Schizophrenia

Correct Answer: C. Schizophreniform disorder

Question 142

Clinical Scenario:

A previously healthy 47-year-old woman comes to the emergency department because of a 2-week history of fatigue, abdominal distention, and vomiting. She drinks 6 beers daily. Physical examination shows pallor and scleral icterus. A fluid wave and shifting dullness are present on abdominal examination. The intravascular pressure in which of the following vessels is most likely to be increased?

Answer Choices:

A. Short gastric vein B. Splenic artery C. Inferior epigastric vein D. Azygos vein

Correct Answer: A. Short gastric vein

Question 143

Clinical Scenario:

A randomized controlled trial was initiated to evaluate a novel DPP-4 inhibitor for blood glucose management in diabetic patients. The study used a commonly prescribed sulfonylurea as the standard of care treatment. 2,000 patients were enrolled in the study with 1,000 patients in each arm. One of the primary outcomes was the development of diabetic nephropathy during treatment. This outcome occurred in 68 patients on the DPP-4 inhibitor and 134 patients on the sulfonylurea. What is the relative risk reduction (RRR) for patients using the DPP-4 inhibitor compared with the sulfonylurea?

Answer Choices:

A. 23% B. 33% C. 49% D. 59%

Correct Answer: C. 49%

Question 144

Clinical Scenario:

A 40-year-old man comes to the physician because of weight gain over the past 3 months. During this period, he has gained 10 kg (22 lb) unintentionally. He also reports decreased sexual desire, oily skin, and sleeping difficulties. There is no personal or family history of serious illness. He has smoked one pack of cigarettes daily for the past 10 years. The patient appears lethargic. His temperature is 37°C (98.6°F), pulse is 80/min, and blood pressure is 150/90 mm Hg. Physical examination shows central obesity, acne, and thin, easily bruisable skin with stretch marks on the abdomen. There is darkening of the mucous membranes and the skin creases. Examination of the muscles shows atrophy and weakness of proximal muscle groups. His serum glucose concentration is 240 mg/dL. Which of the following findings would most likely be present on imaging?

Answer Choices:

A. Pituitary microadenoma B. Decreased thyroid size C. Multiple kidney cysts D. Adrenal carcinoma

Correct Answer: A. Pituitary microadenoma

Question 145

Clinical Scenario:

A 26-year-old African-American woman comes to the physician because of a 4-day history of a nonproductive cough and chest pain. The pain is sharp and worse when she breathes deeply. During this time, she has also had two episodes of hematuria. Over the past 6 months, she has had intermittent pain, stiffness, and swelling in her fingers and left knee. She had two miscarriages at age 22 and 24. Her only medication is minocycline for acne vulgaris. Her temperature is 38.1°C (100.6°F), pulse is 75/min, and blood pressure is 138/85 mm Hg. Physical examination shows an erythematous rash on her face. There is mild tenderness over the metacarpophalangeal joints bilaterally with no warmth or erythema. Further evaluation of this patient is most likely to show which of the following findings?

Answer Choices:

A. Anti-histone antibodies B. Low serum levels of C3 and C4 C. Erosions of the metacarpophalangeal joints D. Cytotoxic glomerular antibodies

Correct Answer: B. Low serum levels of C3 and C4

Question 146

Clinical Scenario:

A 9-year-old boy is brought to the physician by his mother because of a 3-day history of fever and bleeding after brushing his teeth. His mother also reports that her son has asked to be picked up early from soccer practice the past few days because of fatigue. He appears pale and ill. His temperature is 38.3°C (101.1°F), pulse is 115/min, and blood pressure is 100/60 mm Hg. The lungs are clear to auscultation. Examination shows a soft, nontender abdomen with no organomegaly. There are several spots of subcutaneous bleeding on the abdomen and shins. Laboratory studies show a hemoglobin concentration of 7 g/dL, a leukocyte count of 2,000/mm³, a platelet count of 40,000/mm³, and a reticulocyte count of 0.2%. Serum electrolyte concentrations are within normal limits. A bone marrow biopsy is most likely to show which of the following findings?

Answer Choices:

A. Sheets of abnormal plasma cells B. Hypocellular fat-filled marrow with RBCs of normal morphology C. Hypercellular, dysplastic bone marrow with ringed sideroblasts D. Infiltration of the marrow with collagen and fibrous tissue

Correct Answer: B. Hypocellular fat-filled marrow with RBCs of normal morphology

Question 147

Clinical Scenario:

A 26-year-old woman comes to the emergency department because of a 3-day history of nausea and vomiting. Her last menstrual period was 9 weeks ago. A urine pregnancy test is positive. Ultrasonography shows an intrauterine pregnancy consistent in size with a 7-week gestation. The hormone that was measured in this patient's urine to detect the pregnancy is also directly responsible for which of the following processes?

Answer Choices:

A. Maintenance of the corpus luteum B. Inhibition of ovulation C. Development of breast tissue D. Inhibition of preterm uterine contractions

Correct Answer: A. Maintenance of the corpus luteum

Question 148

Clinical Scenario:

A 31-year-old man comes to the physician because of several months of recurrent abdominal pain and diarrhea. Six months ago, he traveled to Lake Superior for a fishing trip with his friends, during which they often ate their day's catch for dinner. Physical examination shows pallor. Laboratory studies show macrocytic anemia with eosinophilia. A peripheral blood smear shows hypochromic red blood cells with megaloblasts and hypersegmented neutrophils. A cestode infection is suspected and a drug is prescribed that kills cestodes by inducing uncontrollable muscle spasm in the parasite. The drug prescribed for this patient most likely acts by which of the following mechanisms of action?

Answer Choices:

A. Increased calcium influx into the sarcoplasm B. Increased sodium efflux from the sarcoplasm C. Increased potassium efflux from the sarcoplasm D. Phosphorylation of adenosine diphosphate

Correct Answer: A. Increased calcium influx into the sarcoplasm

Question 149

Clinical Scenario:

A 40-year-old, gravida 2, nulliparous woman, at 14 weeks' gestation comes to the physician because of a 6-hour history of light vaginal bleeding and lower abdominal discomfort. Eight months ago she had a spontaneous abortion at 10 weeks' gestation. Her pulse is 92/min, respirations are 18/min, and blood pressure is 134/76 mm Hg. Abdominal examination shows no tenderness or masses; bowel sounds are normal. On pelvic examination, there is old blood in the vaginal vault and at the closed cervical os. The uterus is larger than expected for the length of gestation and there are bilateral adnexal masses. Serum β -hCG concentration is 120,000 mIU/ml. Which of the following is the most appropriate next step in management?

Answer Choices:

A. Transvaginal ultrasound B. Chorionic villus sampling C. Thyroid function tests D. Fetal Doppler ultrasound

Correct Answer: A. Transvaginal ultrasound

Question 150

Clinical Scenario:

A 51-year-old woman presents to the emergency department with a 2-day history of bilateral lower extremity swelling. She says that her legs do not hurt, but she noticed she was gaining weight and her legs were becoming larger. Her past medical history is significant for morbid obesity, hypertension, and hypercholesterolemia. She says the swelling started after she was recently started on a new medication to help her blood pressure, but she does not remember the name of the medication. Which of the following is the most likely the mechanism of action for the drug that was prescribed to this patient?

Answer Choices:

A. Inhibition of calcium channels B. Inhibition of hormone receptor C. Potassium-sparing diuretic D. Potassium-wasting diuretic

Correct Answer: A. Inhibition of calcium channels

Question 151

Clinical Scenario:

A 44-year-old woman presents to the outpatient infectious disease clinic. She has a known history of HIV, well-controlled on HAART for the past 8 years. She currently has no additional significant medical conditions. She feels well and a physical examination is within normal limits. She denies any current tobacco use, alcohol use, or illicit drug use, although she has a history of heroin use (injection). Her vital signs include: temperature, 36.7°C (98.0°F); blood pressure, 126/74 mm Hg; heart rate, 87/min; and respiratory rate, 17/min. She has no complaints and is up to date on all of her vaccinations and preventative care. Which of the following malignancies can be seen and is often associated with AIDS?

Answer Choices:

A. Colonic adenocarcinoma B. Kaposi's sarcoma C. Thymomas D. Malignant melanoma

Correct Answer: B. Kaposi's sarcoma

Question 152

Clinical Scenario:

A 3200-g (7.1-lb) female newborn is delivered at 38 weeks' gestation to a 24-year-old woman. The mother had regular prenatal visits throughout the pregnancy. The newborn's blood pressure is 53/35 mm Hg. Examination in the delivery room shows clitoromegaly and posterior labial fusion. One day later, serum studies show: Na⁺

131 mEq/L K⁺ 5.4 mEq/L Cl⁻ 102 mEq/L Urea nitrogen 15 mg/dL Creatinine 0.8 mg/dL Ultrasound of the abdomen and pelvis shows a normal uterus and ovaries. Further evaluation of the newborn is most likely to show which of the following findings?"

Answer Choices:

A. Decreased dehydroepiandrosterone B. Increased 17-hydroxyprogesterone C. Increased corticosterone D. Decreased renin activity

Correct Answer: B. Increased 17-hydroxyprogesterone

Question 153

Clinical Scenario:

A 28-year-old man presents to the clinic with increasing shortness of breath, mild chest pain at rest, and fatigue. He normally lives a healthy lifestyle with moderate exercise and an active social life, but recently he has been too tired to do much. He reports that he is generally healthy and on no medications but did have a 'cold' 2 weeks ago. He does not smoke, besides occasional marijuana with friends, and only drinks socially. His father has hypertension, hyperlipidemia, and lung cancer after a lifetime of smoking, and his mother is healthy. He also has one older brother with mild hypertension. His pulse is 104/min, the respiratory rate 23/min, the blood pressure 105/78 mm Hg, and the temperature 37.1°C (98.8°F). On physical examination, he is ill-appearing and has difficulty completing sentences. On auscultation he has a third heart sound, and his point of maximal impact is displaced laterally. He has 2+ pitting edema of the lower extremities up to the knees. An ECG is obtained and shows premature ventricular complexes and mildly widened QRS complexes. An echocardiogram is also performed and shows global hypokinesis with a left ventricle ejection fraction of 39%. Of the following, what is the most likely cause of his symptoms?

Answer Choices:

A. Acute myocardial infarction B. Unstable angina C. Coxsackievirus infection D. Cocaine abuse

Correct Answer: C. Coxsackievirus infection

Question 154

Clinical Scenario:

During the course of investigation of a suspected abdominal aortic aneurysm in a 57-year-old woman, a solid 6 × 5 cm mass is detected in the right kidney. The abdominal aorta reveals no abnormalities. The patient is feeling well and has no history of any serious illness or medication usage. She is a 25-pack-year smoker. Her vital signs are within normal limits. Physical examination reveals no abnormalities. Biopsy of the mass shows renal cell carcinoma. Contrast-enhanced CT scan indicates no abnormalities involving contralateral kidney, lymph nodes, lungs, liver, bone, or brain. Which of the following treatment options is the most appropriate next step in the management of this patient?

Answer Choices:

A. Interferon- α (IFN- α) B. Interleukin 2 (IL-2) C. Nephrectomy D. Radiation

Correct Answer: C. Nephrectomy

Question 155

Clinical Scenario:

A 36-year-old woman comes to the physician for evaluation of unintentional weight gain of 5.5 kg (12.2 lb) and irregular menstrual cycles over the past 2 months. She does not take any medications. Her blood pressure is 155/85 mm Hg. Physical examination shows central obesity, hyperpigmentation of the palmar creases, and violaceous scarring of the abdomen. Early morning serum cortisol levels are elevated and serum adrenocorticotropic hormone (ACTH) is within the reference range after a low-dose dexamethasone suppression test. A high-dose dexamethasone suppression test shows suppression of ACTH. Further evaluation is most likely to show which of the following findings?

Answer Choices:

A. Benign adenoma of the adrenal medulla B. Nodular hypertrophy of the zona reticularis C. Bilateral hyperplasia of the zona fasciculata D. Unilateral carcinoma of the adrenal cortex

Correct Answer: C. Bilateral hyperplasia of the zona fasciculata

Question 156

Clinical Scenario:

A study is performed to determine the prevalence of a particular rare fungal pneumonia. A sample population of 100 subjects is monitored for 4 months. Every month, the entire population is screened and the number of new cases is recorded for the group. The data from the study are given in the table below:

Time point	New cases of fungal pneumonia
t = 0 months	10
t = 1 months	4
t = 2 months	2
t = 3 months	5
t = 4 months	4

Which of the following is correct regarding the prevalence of this rare genetic condition in this sample population?

Answer Choices:

A. The prevalence at time point 2 months is 2%. B. The prevalence at time point 3 months is 11%. C. The prevalence and the incidence at time point 2 months are equal. D. The prevalence at the conclusion of the study is 25%.

Correct Answer: D. The prevalence at the conclusion of the study is 25%.

Question 157

Clinical Scenario:

A previously healthy 14-year-old boy is brought to the physician for evaluation because of loss of appetite, sleeplessness, and extreme irritability for 3 weeks. He recently quit the school's football team after missing

many practices. He has also been avoiding his family and friends because he is not in the mood to see them but admits that he is lonely. He has not left his room for 2 days, which prompted his father to bring him to the physician. He has no medical conditions and does not take any medications. He does not drink alcohol or use recreational drugs. While the father is in the waiting room, mental status examination is conducted, which shows a constricted affect. Cognition is intact. He says that he would be better off dead and refuses to be treated. He says he wants to use his father's licensed firearm to "end his misery" over the weekend when his parents are at church. Which of the following is the most appropriate next step in management?

Answer Choices:

A. Involuntary hospitalization after informing the parents B. Reassure the patient that he will feel better C. Begin paroxetine therapy D. Start outpatient psychotherapy

Correct Answer: A. Involuntary hospitalization after informing the parents

Question 158

Clinical Scenario:

A 42-year-old man presents with unremitting diarrhea that has lasted for 2 weeks. He describes his bowel movements as watery, non-bloody, foul-smelling, and greasy. He also has cramping abdominal pain associated with the diarrhea. He says that his symptoms started right after he returned from a father-son camping trip to the mountains. His son has similar symptoms. His vital signs include: pulse 78/min, respiratory rate 15/min, temperature 37.2°C (99.0°F), and blood pressure 120/70 mm Hg. A stool sample is obtained and microscopic analysis is significant for the findings shown in the image below. Which of the following pathogens is most likely responsible for this patient's condition?

Answer Choices:

A. *Campylobacter jejuni* B. *Clostridium difficile* C. *Giardia lamblia* D. *Yersinia enterocolitica*

Correct Answer: C. *Giardia lamblia*

Question 159

Clinical Scenario:

A 30-year-old woman presents to her primary care provider with blood in her urine and pain in her left flank. She has a 5-year history of polycystic ovarian syndrome managed with oral contraceptives and metformin. She is single and is not sexually active and denies a history of kidney stones or abdominal trauma. She has a 15-pack-year smoking history but denies the use of other substances. Her family history is significant for fatal lung cancer in her father at age 50, who also smoked, and recently diagnosed bladder cancer in her 45-year-old brother, who never smoked. On review of systems, she denies weight loss, fever, fatigue, paresthesia, increased pain with urination, or excessive bleeding or easy bruising. She is admitted to the hospital for a workup and observation. Her vital signs and physical exam are within normal limits. A urine pregnancy test is negative. PT is 14 sec and PTT is 20 sec. The rest of the laboratory results including von Willebrand factor activity and

lupus anticoagulant panel are pending. A CT angiogram is ordered and is shown in the picture. What is indicated at this time to prevent a potential sequela of this patient's condition?

Answer Choices:

A. Surgery B. Administer heparin C. Thrombectomy D. Administer warfarin

Correct Answer: B. Administer heparin

Question 160

Clinical Scenario:

A 23-year-old woman is admitted to the inpatient psychiatry unit after her boyfriend reported she was "acting funny and refusing to talk." The patient's boyfriend states that he came home from work and found the patient sitting up in bed staring at the wall. When he said her name or waved his hand in front of her, she did not respond. When he tried to move her, she would remain in whatever position she was placed. The patient's temperature is 99°F (37.2°C), blood pressure is 122/79 mmHg, pulse is 68/min, and respirations are 12/min with an oxygen saturation of 98% O₂ on room air. During the physical exam, the patient is lying on the bed with her left arm raised and pointing at the ceiling. She resists any attempt to change her position. The patient remains mute and ignores any external stimuli. The patient's medical history is significant for depression. She was recently switched from phenelzine to fluoxetine. Which of the following is the best initial therapy?

Answer Choices:

A. Benztropine B. Electroconvulsive therapy C. Haloperidol D. Lorazepam

Correct Answer: D. Lorazepam

Question 161

Clinical Scenario:

A 52-year-old man comes to the physician for a routine medical check-up. The patient feels well. He has hypertension, type 2 diabetes mellitus, and recurrent panic attacks. He had a myocardial infarction 3 years ago. He underwent a left inguinal hernia repair at the age of 25 years. A colonoscopy 2 years ago was normal. He works as a nurse at a local hospital. He is married and has two children. His father died of prostate cancer at the age of 70 years. He had smoked one pack of cigarettes daily for 25 years but quit following his myocardial infarction. He drinks one to two beers on the weekends. He has never used illicit drugs. Current medications include aspirin, atorvastatin, lisinopril, metoprolol, fluoxetine, metformin, and a multivitamin. He appears well-nourished. Temperature is 36.8°C (98.2°F), pulse is 70/min, and blood pressure is 125/75 mm Hg. Lungs are clear to auscultation. Cardiac examination shows a high-frequency, mid-to-late systolic murmur that is best heard at the apex. The abdomen is soft and nontender. The remainder of the physical examination shows no abnormalities. Which of the following is the most likely diagnosis?

Answer Choices:

A. Pulmonary valve regurgitation B. Tricuspid valve stenosis C. Pulmonary valve stenosis D. Mitral valve prolapse

Correct Answer: D. Mitral valve prolapse

Question 162

Clinical Scenario:

A 37-year-old man presents to the physician because of dysphagia and regurgitation for the past 5 years. In recent weeks, it has become very difficult for him to ingest solid or liquid food. He has lost 3 kg (6 lb) during this time. He was admitted to the hospital last year because of pneumonia. Three years ago, he had an endoscopic procedure which partially improved his dysphagia. He takes amlodipine and nitroglycerine before meals. His vital signs are within normal limits. BMI is 19 kg/m². Physical examination shows no abnormalities. A barium swallow X-ray is shown. Which of the following patterns of esophageal involvement is the most likely cause of this patient's condition?

Answer Choices:

A. Abnormal esophageal contraction with deglutition lower esophageal sphincter relaxation B. Absent peristalsis and impaired lower esophageal sphincter relaxation C. Poor pharyngeal propulsion and upper esophageal sphincter obstruction D. Severely weak peristalsis and patulous lower esophageal sphincter

Correct Answer: B. Absent peristalsis and impaired lower esophageal sphincter relaxation

Question 163

Clinical Scenario:

A 25-year-old woman comes to the physician because of recurrent episodes of reddish discoloration of her urine. She also has a 3-month history of intermittent abdominal pain, yellowish discoloration of the skin and eyes, and fatigue. Physical examination shows pallor and scleral icterus. The spleen is not palpable. Her hemoglobin concentration is 7.8 g/dL, leukocyte count is 2,000/mm³, and platelet count is 80,000/mm³. Serum LDH and unconjugated bilirubin concentrations are elevated. Addition of a serum containing anti-human globulins to a blood sample shows no agglutination. A urine dipstick shows blood; urinalysis shows no RBCs. A CT scan of the abdomen shows a thrombus in a distal branch of the superior mesenteric vein. Which of the following is the most likely cause of this patient's condition?

Answer Choices:

A. Activation and consumption of platelets and coagulation factors B. Absence of protective factors against destruction by complement C. Formation of IgG antibodies against glycoprotein IIb/IIIa D. Replacement of a single amino acid in a β -globin chain "

Correct Answer: B. Absence of protective factors against destruction by complement

Question 164

Clinical Scenario:

An investigator is studying the activity of N-terminal peptidase in eukaryotes. Sulfur-containing amino acids are radiolabeled and isolated using ³⁵S. During translation of a non-mitochondrial human genome, some of the radiolabeled amino acids bind to the aminoacyl, peptidyl, and exit sites of a eukaryotic ribosome but others bind only to the peptidyl and exit sites. Only the radiolabeled amino acids that do not bind to the ribosomal aminoacyl-site can be excised by the N-terminal peptidase. Which of the following best describes the anticodon sequence of the transfer RNA charged by the amino acid target of the N-terminal peptidase?

Answer Choices:

A. 5'-UCA-3' B. 5'-CAU-3' C. 5'-ACA-3' D. 5'-ACU-3'

Correct Answer: B. 5'-CAU-3'

Question 165

Clinical Scenario:

A 5-year-old non-verbal child with a history of autism is brought into the emergency department by his grandmother. The patient's grandmother is concerned her grandchild is being abused at home. The patient lives in an apartment with his mother, step-father, and two older brothers in low-income housing. The department of social services has an open case regarding this patient and his family. The patient is afebrile. His vital signs include: blood pressure 97/62 mm Hg, pulse 175/min, respiratory rate 62/min. Physical examination reveals a malnourished and dehydrated child in dirty and foul-smelling clothes. Which one of the following people is most likely abusing this patient?

Answer Choices:

A. Mother B. Step-father C. Brother D. Neighbor

Correct Answer: A. Mother

Question 166

Clinical Scenario:

A 56-year-old woman presents to her primary care physician complaining of heartburn, belching, and epigastric pain that is aggravated by coffee and fatty foods. She states that she has recently been having difficulty swallowing in addition to her usual symptoms. What is the most appropriate next step in management of this patient?

Answer Choices:

A. Nissen fundoplication B. Lifestyle changes - don't lie down after eating; avoid spicy foods; eat small servings C. Trial of an H₂ receptor antagonist D. Upper endoscopy

Correct Answer: D. Upper endoscopy

Question 167

Clinical Scenario:

A 62-year-old woman presents to her primary care physician for a routine physical exam. The patient has no specific complaints but does comment on some mild weight gain. She reports that she recently retired from her job as a math teacher and has taken up hiking. Despite the increase in activity, she believes her pants have become "tighter." She denies headaches, urinary symptoms, or joint pains. She has a history of hypertension, type 2 diabetes, and rheumatoid arthritis. Her medications include aspirin, lisinopril, rovastatin, metformin, and methotrexate. She takes her medications as prescribed and is up to date with her vaccinations. A colonoscopy two years ago and a routine mammography last year were both normal. The patient's last menstrual period was 10 years ago. The patient has a father who died of colon cancer at 71 years of age and a mother who has breast cancer. Her temperature is 98.7°F (37°C), blood pressure is 132/86 mmHg, pulse is 86/min, respirations are 14/min and oxygen saturation is 98% on room air. Physical exam is notable for a mildly distended abdomen and a firm and non-mobile right adnexal mass. What is the next step in the management of this patient?

Answer Choices:

A. Abdominal MRI B. CA-125 level C. Exploratory laparotomy and debulking D. Pelvic ultrasound

Correct Answer: D. Pelvic ultrasound

Question 168

Clinical Scenario:

A 26-year-old G1P0 woman is brought to the emergency room by her spouse for persistently erratic behavior. Her spouse reports that she has been sleeping > 1 hour a night, and it sometimes seems like she's talking to herself. She has maxed out their credit cards on baby clothes. The patient's spouse reports this has been going on for over a month. Since first seeing a physician, she has been prescribed multiple first and second generation antipsychotics, but the patient's spouse reports that her behavior has failed to improve. Upon examination, the patient is speaking rapidly and occasionally gets up to pace the room. She reports she is doing "amazing," and that she is "so excited for the baby to get here because I'm going to be the best mom." She denies illicit drug use, audiovisual hallucinations, or suicidal ideation. The attending psychiatrist prescribes a class of medication the patient has not yet tried to treat the patient's psychiatric condition. In terms of this new medication, which of the following is the patient's newborn most likely at increased risk for?

Answer Choices:

A. Caudal regression syndrome B. Ototoxicity C. Renal defects D. Right ventricular atrialization

Correct Answer: D. Right ventricular atrialization

Question 169

Clinical Scenario:

A 61-year-old woman presents to her primary care physician complaining of left-sided facial pain that started yesterday. She describes the pain as stinging, burning, and constant. It does not worsen with jaw movement or chewing. Her past medical history includes hyperlipidemia and multiple sclerosis (MS), and she had chickenpox as a child but received a shingles vaccination last year. Medications include simvastatin and glatiramer acetate. The patient's last MS flare was 5 weeks ago, at which time she received a prednisone burst with taper. At this visit, her temperature is 99.9 °F (37.7°C), blood pressure is 139/87 mmHg, pulse is 82/min, and respirations are 14/min. On exam, there is no rash or skin change on either side of the patient's face. Gentle palpation of the left cheek and mandible produce significant pain, but there is full range of motion in the jaw. Which of the following medications is the most likely to prevent long-term persistence of this patient's pain?

Answer Choices:

A. Carbamazepine B. Amitriptyline C. Oral acyclovir D. Gabapentin

Correct Answer: C. Oral acyclovir

Question 170

Clinical Scenario:

A 32-year-old woman comes to the office for a regular follow-up. She was diagnosed with type 2 diabetes mellitus 4 years ago. Her last blood test showed a fasting blood glucose level of 6.6 mmol/L (118.9 mg/dL) and HbA1c of 5.1%. No other significant past medical history. Current medications are metformin and a daily multivitamin. No significant family history. The physician wants to take her blood pressure measurements, but the patient states that she measures it every day in the morning and in the evening and even shows him a blood pressure diary with all the measurements being within normal limits. Which of the following statements is correct?

Answer Choices:

A. The physician should not measure the blood pressure in this patient and should simply make a note in a record showing the results from the patient's diary. B. The physician has to measure the patient's blood pressure because it is a standard of care for any person with diabetes mellitus who presents for a check-up. C. The physician should not measure the blood pressure in this patient because the local standards of care in the physician's office differ from the national standards of care so measurements of this patient's blood pressure can not be compared to diabetic D. Assessment of blood pressure only needs to be done at the initial visit; it is not necessary to measure blood pressure in this patient at any follow-up appointments.

Correct Answer: B. The physician has to measure the patient's blood pressure because it is a standard of care for any person with diabetes mellitus who presents for a check-up.

Question 171

Clinical Scenario:

A 26-year-old man is brought to the emergency department because of abdominal pain, dizziness, shortness of breath, and swelling and pruritus of the lips, tongue, and throat for 1 hour. The symptoms began minutes after he started eating a lobster dinner. It is determined that his symptoms are due to surface crosslinking of IgE. This immunologic event most likely caused the release of which of the following?

Answer Choices:

A. Tryptase B. Cathepsin C. Bradykinin D. Interferon gamma "

Correct Answer: A. Tryptase

Question 172

Clinical Scenario:

A 25-year-old G2P1 woman at 28 weeks estimated gestational age presents with questions on getting epidural anesthesia for her upcoming delivery. She has not received any prenatal care until now. Her previous pregnancy was delivered safely at home by an unlicensed midwife, but she would like to receive an epidural for this upcoming delivery. Upon inquiry, she admits that she desires a "fully natural experience" and has taken no supplements or shots during or after her 1st pregnancy. Her 1st child also did not receive any post-delivery injections or vaccinations but is currently healthy. The patient has an A (-) negative blood group, while her husband has an O (+) positive blood group. Which of the following should be administered immediately in this patient to prevent a potentially serious complication during delivery?

Answer Choices:

A. Folic acid B. Vitamin K C. Anti-RhO(D) immunoglobulin D. Vitamin D

Correct Answer: C. Anti-RhO(D) immunoglobulin

Question 173

Clinical Scenario:

A 44-year-old man presents for a checkup. The patient says he has to urinate quite frequently but denies any dysuria or pain on urination. Past medical history is significant for diabetes mellitus type 2 and hypertension, both managed medically, as well as a chronic mild cough for the past several years. Current medications are metformin, aspirin, rosuvastatin, captopril, and furosemide. His vital signs are an irregular pulse of 74/min, a respiratory rate of 14/min, a blood pressure of 130/80 mm Hg, and a temperature of 36.7°C (98.0°F). His BMI is 32 kg/m². On physical examination, there are visible jugular pulsations present in the neck bilaterally. Laboratory findings are significant for the following: Glycated Hemoglobin (Hb A1c) 7.5% Fasting Blood Glucose 120 mg/dL Serum Electrolytes Sodium 138 mEq/L Potassium 3.9 mEq/L Chloride 101 mEq/L Serum

Creatinine 1.3 mg/dL Blood Urea Nitrogen 18 mg/dL Which of the following is the next best step in the management of this patient?

Answer Choices:

A. Stop metformin. B. Replace captopril with valsartan. C. Start rosiglitazone. D. Start exenatide.

Correct Answer: D. Start exenatide.

Question 174

Clinical Scenario:

A 16-year-old girl is brought to the emergency department by her parents because of fever, vomiting, rash, and worsening confusion since this morning. On questioning, her mother reports that her last menstrual period was 1 week ago and that she recently started using tampons. She appears lethargic and is only oriented to person. Her temperature is 40.4°C (104.7°F), pulse 174/minute, and blood pressure is 62/44 mm Hg. Examination shows oropharyngeal hyperemia and diffuse macular erythroderma. Which of the following is the most likely cause of this patient's condition?

Answer Choices:

A. Erythrogenic toxin production B. Lipooligosaccharide expression C. Unregulated B cell proliferation D. Polyclonal T cell activation

Correct Answer: D. Polyclonal T cell activation

Question 175

Clinical Scenario:

A 15-year-old girl is brought into the clinic for a second opinion by her mother. She was recently diagnosed with alopecia areata after having presented to her family doctor with a 2-month history of noticeable bald patches. The mother was concerned because the treatment involved topical steroids. The patient is noted to have started high school earlier this year and has an attention-deficit/ hyperactivity disorder that is treated with methylphenidate. The patient is afebrile and her vital signs are within normal limits. Physical examination reveals a moderately distressed young female playing with her hair. She has very short nails on both hands and patches of hair loss on the scalp. Which of the following is the most appropriate initial management for this patient's condition?

Answer Choices:

A. Behavioral therapy B. Selective serotonin reuptake inhibitors C. Antipsychotics D. Lithium

Correct Answer: A. Behavioral therapy

Question 176

Clinical Scenario:

An 8-year-old girl is brought to the physician for a well-child examination. Since the age of 2 years, she has had multiple fractures after minor trauma. During the past year, she has fractured the left humerus and right clavicle after falls. Her father also has a history of recurrent fractures. She is at the 5th percentile for height and 20th percentile for weight. Vital signs are within normal limits. Physical examination shows increased convexity of the thoracic spine. Forward bend test demonstrates asymmetry of the thoracolumbar region. There is a curvature of the tibias bilaterally, and the left leg is 2 cm longer than the right. There is increased mobility of the joints of the upper and lower extremities. Which of the following is the most likely additional finding?

Answer Choices:

A. Dislocated lens B. Hearing impairment C. Widely spaced permanent teeth D. Cerebral berry aneurysm

Correct Answer: B. Hearing impairment

Question 177

Clinical Scenario:

A 33-year-old woman presents to the clinic complaining of a 9-month history of weight loss, fatigue, and a general sense of malaise. She additionally complains of an unusual sensation in her chest upon rapidly rising from a supine to a standing position. Current vitals include a temperature of 36.8°C (98.2°F), pulse of 72/min, blood pressure of 118/63 mm Hg, and a respiratory rate of 15/min. Her BMI is 21 kg/m². Auscultation demonstrates an early-mid diastole low-pitched sound at the apex of the heart. A chest X-ray reveals a poorly demarcated abnormality in the heart and requires CT imaging for further analysis. What would most likely be seen on CT imaging?

Answer Choices:

A. Tumor within the right atria B. Fistula between the right and left atria C. Normal cardiac imaging D. Tumor within the left atria

Correct Answer: D. Tumor within the left atria

Question 178

Clinical Scenario:

A 33-year-old woman presents to her primary care physician for non-bloody nipple discharge. She states that it has been going on for the past month and that it sometimes soils her shirt. The patient drinks 2 to 3 alcoholic beverages per day and smokes 1 pack of cigarettes per day. She is currently seeking mental health treatment with an outpatient psychiatrist after a recent hospitalization for auditory hallucinations. Her psychiatrist prescribed her a medication that she can not recall. Otherwise, she complains of headaches that occur frequently. Her temperature is 98.6°F (37.0°C), blood pressure is 137/68 mmHg, pulse is 70/min, respirations

are 13/min, and oxygen saturation is 98% on room air. Physical exam is notable for bilateral galactorrhea that can be expressed with palpation. Which of the following is the best next step in management?

Answer Choices:

A. Discontinuation of current psychiatric medications B. Mammography C. TSH level D. Ultrasound and biopsy

Correct Answer: C. TSH level

Question 179

Clinical Scenario:

A 25-year-old man is brought to the emergency department 30 minutes after he was involved in a motorcycle collision. He was not wearing a helmet. Physical examination shows left periorbital ecchymosis. A CT scan of the head shows a fracture of the greater wing of the left sphenoid bone with compression of the left superior orbital fissure. Physical examination of this patient is most likely to show which of the following findings?

Answer Choices:

A. Decreased sense of smell B. Numbness of the left cheek C. Preserved left lateral gaze D. Absent left corneal reflex

Correct Answer: D. Absent left corneal reflex

Question 180

Clinical Scenario:

A 34-year-old woman, gravida 2, para 2, is admitted to the hospital because of shortness of breath and fatigue 2 weeks after delivery of a full-term female newborn. She has no history of major medical illness. Cardiac examination on admission shows an S3 gallop and a grade 2/6 holosystolic murmur heard best at the apex. Treatment is initiated with intravenous furosemide and captopril. Her symptoms resolve, and 3 weeks later, cardiac examination shows no murmur. Which of the following is the most likely explanation for the initial auscultation findings?

Answer Choices:

A. Mitral annular dilatation B. Myxomatous mitral valve degeneration C. Mitral valve leaflet fibrosis D. Mitral annular calcification

Correct Answer: A. Mitral annular dilatation

Question 181

Clinical Scenario:

A 32-year-old woman has painless abdominal distention 30 hours after an elective repeat cesarean section. She has passed stool once. She has nausea, but no vomiting. Antibiotic prophylaxis was given prior to delivery. She is ambulating without difficulty. Her temperature is 36.7°C (98.1°F), pulse is 95/min, respiratory rate is 17/min, and blood pressure is 100/70 mm Hg. The lungs are clear to auscultation. The abdominal examination shows symmetric distention, a normal surgical scar, absent bowel sounds, and tympanic percussion without tenderness. An abdominal X-ray reveals diffuse bowel distention. The laboratory studies show the following: Laboratory test Hemoglobin 13 g/dL Leukocyte count 11500/mm³ Neutrophils 70% Lymphocytes 24% Serum Na⁺ 137 mEq/L K⁺ 3.2 mEq/L Cl⁻ 104 mEq/L HCO₃⁻ 23 mEq/L Urea nitrogen 22 mg/dL Creatinine 0.8 mg/dL Which of the following is the most appropriate initial management?

Answer Choices:

A. Broad-spectrum antibiotics B. Intravenous fluids and electrolytes C. Laparoscopic exploration D. Laparotomy

Correct Answer: B. Intravenous fluids and electrolytes

Question 182

Clinical Scenario:

A 50-year-old woman comes to the physician for the evaluation of fatigue over the past 6 months. During this period, the patient has also had a 5 kg (11-lb) weight loss. She has a history of Hashimoto thyroiditis. She is sexually active with her husband only. She does not smoke. She drinks one glass of wine per day. She does not use illicit drugs. Her only medication is levothyroxine. Temperature is 37°C (98.6°F), pulse is 70/min, and blood pressure is 110/70 mm Hg. Abdominal examination shows tenderness in the right upper quadrant with no rebound or guarding. Laboratory studies show a serum alanine aminotransferase level of 190 U/L, serum aspartate aminotransferase level of 250 U/L, and serum total bilirubin level of 0.6 mg/dL. Liver biopsy shows plasma cell infiltration and areas of periportal piecemeal necrosis. Further evaluation of this patient is most likely to show which of the following findings?

Answer Choices:

A. Positive anti-smooth muscle antibodies B. Positive anti-mitochondrial antibodies C. Positive HBV surface antigen D. Elevated serum transferrin saturation

Correct Answer: A. Positive anti-smooth muscle antibodies

Question 183

Clinical Scenario:

A previously healthy 24-year-old man is brought to the emergency department 30 minutes after an episode of loss of consciousness. He was standing in line at a bus stop when he suddenly became tense, fell down, and lost consciousness; this was followed by 4 minutes of violent jerky movements of his arms and legs. He was confused after the episode. He has no recollection of the event or its immediate aftermath. On arrival, he is alert and oriented to time, place, and person. His temperature is 37.7°C (99.4°F), pulse is 98/min, and blood pressure

is 130/70 mm Hg. Physical examination shows blood in the mouth. Neurologic examination shows no focal findings. A CT scan of the head shows no abnormalities. Further evaluation of this patient is most likely to show which of the following laboratory findings?

Answer Choices:

A. Increased serum calcium B. Increased serum sodium C. Reduced serum bicarbonate D. Increased serum magnesium

Correct Answer: C. Reduced serum bicarbonate

Question 184

Clinical Scenario:

A 42-year-old woman is seen by her primary care physician for her annual checkup. She has no current concerns and says that she has been healthy over the last year except for a bout of the flu in December. She has no significant past medical history and is not currently taking any medications. She has smoked 1 pack per day since she was 21 and drinks socially with her friends. Her family history is significant for prostate cancer in her dad when he was 51 years of age and ovarian cancer in her paternal aunt when she was 41 years of age. Physical exam reveals a firm, immobile, painless lump in the upper outer quadrant of her left breast as well as 2 smaller nodules in the lower quadrants of her right breast. Biopsy of these lesions shows small, atypical, glandular, duct-like cells with stellate morphology. Which of the following pathways is most likely abnormal in this patient?

Answer Choices:

A. Base excision repair B. Homologous recombination C. Mismatch repair D. Non-homologous end joining

Correct Answer: B. Homologous recombination

Question 185

Clinical Scenario:

A 22-year-old man comes to the physician because of a 3-week history of abdominal pain, loose, non-bloody stools, and intermittent nausea. He also reports intermittent fever. He has not had vomiting, tenesmus, or rectal pain. He has no history of serious illness and takes no medications. His vital signs are within normal limits. Rectal exam is unremarkable. Laboratory studies show a leukocyte count of 15,200/mm³ and an erythrocyte sedimentation rate of 44 mm/h. Test of the stool for occult blood and stool studies for infection are negative. A CT scan of the abdomen shows mural thickening and surrounding fat stranding of discrete regions of the terminal ileum and transverse colon. A colonoscopy is performed and biopsy specimens of the affected areas of the colon are taken. Which of the following findings is most specific for this patient's most likely diagnosis?

Answer Choices:

A. Neutrophilic inflammation of the crypts B. Inflammation of the terminal ileum C. Intranuclear and cytoplasmic inclusion bodies D. Non-caseating granulomas

Correct Answer: D. Non-caseating granulomas

Question 186

Clinical Scenario:

A 45-year-old woman comes to the clinic for complaints of abdominal pain and repeated watery stools for the past 2 days. She has a history of bowel complaints for the past 2 years consisting of periods of intermittent loose stools followed by the inability to make a bowel movement. Her past medical history is significant for diabetes controlled with metformin. She denies any abnormal oral intake, weight loss, fever, nausea/vomiting, or similar symptoms in her family. When asked to describe her stool, she reports that "it is just very watery and frequent, but no blood." The physician prescribes a medication aimed to alleviate her symptoms. What is the most likely mechanism of action of this drug?

Answer Choices:

A. D2 receptor antagonist B. PGE1 analog C. Substance P antagonist D. mu-opioid receptor agonist

Correct Answer: D. mu-opioid receptor agonist

Question 187

Clinical Scenario:

A 46-year-old woman comes to the physician for a 6-month history of worsening bronchial asthma control. Before this issue began, she only used her salbutamol inhaler once a day. Now, she has to use it multiple times daily and also reports frequent nighttime awakening. Seven months ago, she moved to an apartment that is damp and has mold on some of the walls. The physician injects 0.1 mL of *Candida albicans* extract on the mid-volar surface of the right arm intradermally. After 48 hours there is a palpable induration of 17 mm. This reaction is most likely a result of release of which of the following substances?

Answer Choices:

A. Tryptase B. Interleukin-10 C. Interferon- γ D. Superoxide anion "

Correct Answer: C. Interferon- γ

Question 188

Clinical Scenario:

A previously healthy 27-year-old man is brought to the emergency department 35 minutes after being involved in a high-speed motor vehicle collision in which he was an unrestrained passenger. He was ambulatory at the accident scene, with stable vital signs and no major external injuries except abrasions to both upper extremities. On arrival, he is alert and oriented. His temperature is 37.3°C (99.1°F), pulse is 88/min, respirations are 14/min, and blood pressure is 128/74 mm Hg. Abdominal examination shows ecchymosis over the upper

abdomen, with tenderness to palpation over the left upper quadrant. There is no guarding or rigidity. Rectal examination is unremarkable. A CT scan of the abdomen with intravenous contrast shows a subcapsular splenic hematoma comprising 8% of the surface area, with no contrast extravasation and minimal blood in the peritoneal cavity. Which of the following is the next best step in management?

Answer Choices:

A. Laparoscopic splenectomy B. Hospitalization and frequent ultrasounds C. Exploratory laparotomy and splenectomy D. Coil embolization of short gastric vessels

Correct Answer: B. Hospitalization and frequent ultrasounds

Question 189

Clinical Scenario:

A 26-year-old woman is referred to a reproduction specialist because of an inability to conceive. She comes with her husband, who was previously examined for causes of male infertility, but was shown to be healthy. The patient has a history of 2 pregnancies at the age of 15 and 17, which were both terminated in the first trimester. She had menarche at the age of 11, and her menstrual cycles began to be regular at the age of 13. Her menses are now regular, but painful and heavy. Occasionally, she notes a mild pain in the lower left quadrant of her abdomen. Her past medical history is also significant for episodes of depression, but she currently denies any depressive symptoms. Current medications are sertraline daily and cognitive-behavioral therapy twice a week. After reviewing her history, the doctor suggests performing an exploratory laparoscopy with salpingoscopy. He explains the flow of the procedure and describes the risks and benefits of the procedure to the patient and her husband. The patient says she understands all the risks and benefits and agrees to undergo the procedure, but her husband disagrees and insists that he should have the final word because his wife is "a mentally unstable woman." Which of the following is correct about the informed consent for the procedure in this patient?

Answer Choices:

A. The patient can make the decision about the treatment herself because she does not show signs of decision-making incapability. B. The decision must be made by both the wife and the husband because of the patient's mental illness. C. Because of the patient's mental disease, the consent should be given by her husband. D. The patient does not have the capacity to make her own decisions because she is taking a psychotropic medication.

Correct Answer: A. The patient can make the decision about the treatment herself because she does not show signs of decision-making incapability.

Question 190

Clinical Scenario:

A 25-year-old man presents to the emergency department complaining of palpitations, lightheadedness, and sweating. He just started working at an investment firm and has been working long hours to make a good impression. Today, he had a dozen cups of espresso to keep himself awake and working. He has never had such

an episode before. His past medical history is unremarkable. His pulse is 150/min, blood pressure is 134/88 mm Hg, respirations are 12/min, and temperature is 36.7°C (98.0°F). ECG shows supraventricular tachycardia with sinus rhythm. Which of the following is the next best step in the management of this patient?

Answer Choices:

A. Adenosine infusion B. Dipping his face in warm water C. Synchronized cardioversion D. Valsalva maneuver

Correct Answer: D. Valsalva maneuver

Question 191

Clinical Scenario:

A 23-year-old man comes to the emergency department with an open wound on his right hand. He states that he got into a bar fight about an hour ago. He appears heavily intoxicated and does not remember the whole situation, but he does recall lying on the ground in front of the bar after the fight. He does not recall any history of injuries but does remember a tetanus shot he received 6 years ago. His temperature is 37°C (98.6°F), pulse is 77/min, and blood pressure is 132/78 mm Hg. Examination shows a soft, nontender abdomen. His joints have no bony deformities and display full range of motion. There is a 4-cm (1.6-in) lesion on his hand with the skin attached only on the ulnar side. The wound, which appears to be partly covered with soil and dirt, is irrigated and debrided by the hospital staff. Minimal erythema and no purulence is observed in the area surrounding the wound. What is the most appropriate next step in management?

Answer Choices:

A. Apposition of wound edges under tension + rifampin B. Surgical treatment with skin graft + tetanus vaccine C. Application of moist sterile dressing + tetanus vaccine D. Surgical treatment with skin flap + ciprofloxacin

Correct Answer: C. Application of moist sterile dressing + tetanus vaccine

Question 192

Clinical Scenario:

A 33-year-old man comes to the emergency department because of a dry mouth and blurred vision for the past 30 minutes. Prior to this, he was on a road trip and started to feel nauseous, dizzy, and fatigued, so his friend gave him a drug that had helped in the past. Physical examination shows dry mucous membranes and dilated pupils. The remainder of the examination shows no abnormalities. Administration of which of the following drugs is most likely to cause a similar adverse reaction in this patient?

Answer Choices:

A. Oxycodone B. Oxybutynin C. Phenylephrine D. Loratadine

Correct Answer: B. Oxybutynin

Question 193

Clinical Scenario:

A 45-year-old homeless man is brought to the emergency department. He was found unconscious at the park. The patient has a past medical history of IV drug abuse, hepatitis C, alcohol abuse, schizophrenia, and depression. He does not receive normal medical follow up or care. His temperature is 102°F (38.9°C), blood pressure is 97/68 mmHg, pulse is 120/min, respirations are 22/min, and oxygen saturation is 98% on room air. Physical exam demonstrates a diffusely distended abdomen that is dull to percussion with a notable fluid wave. The abdominal exam causes the patient to contract his extremities. Cardiac and pulmonary exam are within normal limits. The patient responds to painful stimuli and smells heavily of alcohol. Which of the following is the best next step in management?

Answer Choices:

A. Cefotaxime B. Ceftriaxone C. Paracentesis D. Ultrasound

Correct Answer: C. Paracentesis

Question 194

Clinical Scenario:

An 83-year-old man with advanced-stage prostate cancer comes to the physician because of a 1-week history of worsening lower back and hip pain. The patient's analgesic regimen includes oxycodone, ibuprofen, and alendronic acid. Physical examination shows localized tenderness over the lumbar spine and right hip. His current pain management requires opioid dose escalation. Which of the following opioid side effects is most likely to remain unaffected by the mechanism underlying this patient's need for a higher drug dose?

Answer Choices:

A. Constipation B. Pruritus C. Nausea D. Respiratory depression

Correct Answer: A. Constipation

Question 195

Clinical Scenario:

A 62-year-old man comes to the physician because of a 2-day history of fever, chills, and flank pain. Five days ago, he was catheterized for acute urinary retention. His temperature is 39.3°C (102.7°F). Physical examination shows right-sided costovertebral angle tenderness. Urine studies show numerous bacteria and WBC casts. Urine culture on blood agar grows mucoid, gray-white colonies. Urine culture on eosin methylene blue agar grows purple colonies with no metallic green sheen. Which of the following is the most likely causal pathogen?

Answer Choices:

A. *Proteus mirabilis* B. *Pseudomonas aeruginosa* C. *Klebsiella pneumoniae* D. *Staphylococcus saprophyticus*

Correct Answer: C. Klebsiella pneumoniae

Question 196

Clinical Scenario:

A 31-year-old G3P1 woman who is at 37 weeks of gestation is brought into the emergency department by her husband after repeated twitching. According to the husband, they were watching TV when she suddenly became unresponsive and her hands and legs started shaking. The episode lasted about 5 minutes and she “seemed to be sleeping afterwards.” Her past medical history is significant for pregnancy induced hypertension. The patient is tired but responsive and denies urinary incontinence, paresthesia, fever, upper respiratory signs, or new medication changes. She denies a history of seizures. Her temperature is 99°F (37°C), blood pressure is 186/97 mmHg, pulse is 96/min, and respirations are 12/min. Physical examination demonstrates a lethargic patient with moderate right upper quadrant tenderness. What is the most appropriate next step for this patient?

Answer Choices:

A. Emergency cesarean section B. Expectant management C. Intravenous infusion of oxytocin D. Intravenous magnesium sulfate

Correct Answer: D. Intravenous magnesium sulfate

Question 197

Clinical Scenario:

A 33-year-old woman presents to her primary care provider for a normal check-up. She reports she has been feeling intermittently fatigued over the past 3 months, but she attributes it to her work as a corporate lawyer and balancing family life. She is otherwise healthy and takes no medications. She was adopted and has no information about her biological family. She has 2 children and has been married for 7 years. She drinks 4-5 glasses of wine per week and does not smoke. Her temperature is 99.2°F (37.3°C), blood pressure is 125/65 mmHg, pulse is 78/min, and respirations are 18/min. On exam, she is well-appearing and in no acute distress. A complete blood count is within normal limits. Additional workup is shown below:

Serum: Na⁺: 139 mEq/L Cl⁻: 99 mEq/L K⁺: 3.9 mEq/L HCO₃⁻: 23 mEq/L BUN: 18 mg/dL Glucose: 110 mg/dL Creatinine: 1.1 mg/dL Ca²⁺: 11.1 mg/dL Parathyroid hormone: 700 pg/mL

Urine: Na⁺: 100 mEq/L/24h Ca²⁺: 100 mg/24h Osmolality: 400 mOsmol/kg H₂O

Which of the following is the most likely underlying cause of this patient's condition?

Answer Choices:

A. Decreased sensitivity to parathyroid hormone B. Impaired tubular calcium reabsorption C. Loss-of-function mutation in the calcium-sensing receptor D. Over-secretion of parathyroid hormone

Correct Answer: C. Loss-of-function mutation in the calcium-sensing receptor

Question 198

Clinical Scenario:

A 39-year-old woman presents to the emergency department with right upper quadrant abdominal discomfort for the past couple of hours. She says that the pain is dull in nature and denies any radiation. She admits to having similar episodes of pain in the past which subsided on its own. Her temperature is 37°C (99.6°F), respirations are 16/min, pulse is 78/min, and blood pressure is 122/98 mm Hg. Physical examination is normal except for diffuse tenderness of her abdomen. She undergoes a limited abdominal ultrasound which reveals a 1.4 cm gallbladder polyp. What is the next best step in the management of this patient?

Answer Choices:

A. Cholecystectomy B. Endoscopic retrograde cholangiopancreatography (ERCP) C. Magnetic resonance cholangiopancreatography (MRCP) D. No further treatment required

Correct Answer: A. Cholecystectomy

Question 199

Clinical Scenario:

A previously healthy 25-year-old woman comes to the physician because of a 1-month history of palpitations that occur on minimal exertion and sometimes at rest. She has no chest discomfort or shortness of breath. She feels nervous and irritable most of the time and attributes these feelings to her boyfriend leaving her 2 months ago. Since then she has started exercising more frequently and taking an herbal weight-loss pill, since which she has lost 6.8 kg (15 lb) of weight. She finds it hard to fall asleep and awakens 1 hour before the alarm goes off each morning. She has been drinking 2 to 3 cups of coffee daily for the past 7 years and has smoked one pack of cigarettes daily for the past 3 years. Her temperature is 37.4°C (99.4°F), pulse is 110/min, respirations are 18/min, and blood pressure is 150/70 mm Hg. Examination shows moist palms. Neurologic examination shows a fine resting tremor of the hands. Deep tendon reflexes are 3+ with a shortened relaxation phase. Which of the following is the most likely cause of this patient's symptoms?

Answer Choices:

A. Coffee consumption B. Hashimoto thyroiditis C. Exogenous hyperthyroidism D. Generalized anxiety disorder "

Correct Answer: C. Exogenous hyperthyroidism

Question 200

Clinical Scenario:

A 26-year-old woman, G1P0, at 22 weeks of gestation presents to the clinic for a prenatal visit. Her recent pregnancy scan shows a single live intrauterine fetus with adequate fetal movements. Facial appearance shows the presence of a cleft lip. The rest of the fetal development is within normal limits. The fetal heart rate is 138/

min. Her prenatal screening tests for maternal serum α -fetoprotein (MSAFP) concentration, pregnancy-associated plasma protein-A (PAPP-A), and free β -human chorionic gonadotropin (β -hCG) are within normal ranges respectively. Her past medical and surgical histories are negative. She is worried about the health of her baby. The baby is at increased risk for which of the following birth defects?

Answer Choices:

A. Respiratory difficulty B. Neural tube abnormalities C. Trisomy 13 D. Ocular abnormalities

Correct Answer: A. Respiratory difficulty

Question 201

Clinical Scenario:

A previously healthy 2-month-old girl is brought to the emergency department because her lips turned blue while passing stools 30 minutes ago. She is at the 40th percentile for length and below the 35th percentile for weight. Pulse oximetry on room air shows an oxygen saturation of 65%, which increases to 76% on administration of 100% oxygen. Physical examination shows perioral cyanosis and retractions of the lower ribs with respiration. Cardiac examination shows a harsh grade 2/6 systolic crescendo-decrescendo murmur heard best at the left upper sternal border. Which of the following is most likely to improve this patient's symptoms?

Answer Choices:

A. Elevation of the lower extremities B. Administration of indomethacin C. Hyperextension of the neck D. Knee to chest positioning

Correct Answer: D. Knee to chest positioning

Question 202

Clinical Scenario:

A 46-year-old woman comes to the physician for a cognitive evaluation. She is an office manager. She has had increasing difficulties with multitasking and reports that her job performance has declined over the past 1 year. On mental status examination, short-term memory is impaired and long-term memory is intact. Laboratory studies, including thyroid-stimulating hormone and vitamin B12, are within the reference range. An MRI of the brain shows generalized atrophy, most pronounced in the bilateral medial temporal lobes and hippocampi. If this patient's condition has a genetic etiology, which of the following alterations is most likely to be found on genetic testing?

Answer Choices:

A. Noncoding hexanucleotide repeats B. Mutation in presenilin 1 C. Expansion of CAG trinucleotide repeat D. Presence of ApoE ϵ 4 allele

Correct Answer: B. Mutation in presenilin 1

Question 203

Clinical Scenario:

A 57-year-old man presents to the emergency department for weight loss and abdominal pain. The patient states that he has felt steadily more fatigued over the past month and has lost 22 pounds without effort. Today, he fainted prompting his presentation. The patient has no significant past medical history. He does have a 33 pack-year smoking history and drinks 4 to 5 alcoholic drinks per day. His temperature is 99.5°F (37.5°C), blood pressure is 100/58 mmHg, pulse is 100/min, respirations are 17/min, and oxygen saturation is 98% on room air. On physical exam, you see a patient who is very thin and appears to be pale. Stool fecal occult blood testing is positive. A CT scan of the abdomen is performed demonstrating a mass in the colon with multiple metastatic lesions scattered throughout the abdomen. The patient is informed of his diagnosis of metastatic colon cancer. When the patient conveys the information to his family he focuses his efforts on discussing the current literature in the field and the novel therapies that have been invented. He demonstrates his likely mortality outcome which he calculated using the results of a large multi-center study. Which of the following is this patient most likely demonstrating?

Answer Choices:

A. Optimism B. Pessimism C. Intellectualization D. Dissociation

Correct Answer: C. Intellectualization

Question 204

Clinical Scenario:

A 40-year-old woman with a recent history of carcinoma of the breast status post mastectomy and adjuvant chemotherapy one week ago presents for follow-up. She reports adequate pain control managed with the analgesic drug she was prescribed. Past medical history is significant for hepatitis C and major depressive disorder. The patient denies any history of smoking or alcohol use but says she is currently using intravenous heroin and has been for the past 10 years. However, she reports that she has been using much less heroin since she started taking the pain medication, which is confirmed by the toxicology screen. Which of the following is the primary mechanism of action of the analgesic drug she was most likely prescribed?

Answer Choices:

A. Mixed agonist-antagonist at opioid receptors B. Pure antagonist at opioid receptors C. Inhibits prostaglandin synthesis D. Pure agonist at the μ -opioid receptor

Correct Answer: A. Mixed agonist-antagonist at opioid receptors

Question 205

Clinical Scenario:

The occupational health department at a hospital implements new safety precautions to prevent laboratory-acquired infections. One of the new precautions includes disinfecting the microbiology laboratory benches with 70% ethanol before and after use. This measure is most likely to be effective in preventing the transmission of which of the following viruses?

Answer Choices:

A. Hepatitis A virus B. Parvovirus C. Polyomavirus D. Herpes simplex virus

Correct Answer: D. Herpes simplex virus

Question 206

Clinical Scenario:

A 27-year-old woman presents for her routine annual examination. She has no complaints. She has a 3-year-old child who was born via normal vaginal delivery with no complications. She had a Pap smear during her last pregnancy and the findings were normal. Her remaining past medical history is not significant, and her family history is also not significant. Recently, one of her close friends was diagnosed with breast cancer at the age of 36, and, after reading some online research, she wants to be checked for all types of cancer. Which of the following statements would be the best advice regarding the most appropriate screening tests for this patient?

Answer Choices:

A. "We should do a Pap smear now. Blood tests are not recommended for screening purposes." B. "You need HPV (human papillomavirus) co-testing only." C. "Yes, you are right to be concerned. Let us do a mammogram and a blood test for CA-125." D. "Your last Pap smear 3 years ago was normal. We can repeat it after 2 more years."

Correct Answer: A. "We should do a Pap smear now. Blood tests are not recommended for screening purposes."

Review Summary

Category: Next Step/Workup **Questions Reviewed:** ____ / 206

Overall Assessment: - Correctly categorized: _ questions - **Miscategorized:** questions - **Ambiguous:** __ questions

General Comments:

Reviewer Name: _____

Date: _____

Credentials: _____